



A comparison study of myoelectric regression performances when estimating different types of joint kinematic data

Eric Cito Becman^{a,*}, Larissa Driemeier^a, Oron Levin^{b,c}, Stephan P. Swinnen^{b,d}, Arturo Forner-Cordero^a

^a Biomechatronics Laboratory, Department of Mechatronics and Mechanical Systems, Polytechnic School, University of São Paulo, São Paulo, 05508-030, São Paulo, Brazil

^b Movement Control and Neuroplasticity Research Group, Department of Movement Sciences, Katholieke Universiteit Leuven, Leuven, 3000, Flemish Brabant, Belgium

^c Department of Health Promotion and Rehabilitation, Lithuanian Sports University, Kaunas, LT-44221, Lithuania

^d Leuven Brain Institute, Katholieke Universiteit Leuven, Leuven, 3000, Flemish Brabant, Belgium

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ABSTRACT

Myoelectric control is used in assistive devices such as powered prostheses or exoskeletons, and advances in this area could greatly improve the quality of life for individuals with motor disabilities. Regression machine learning models have been studied to provide a more natural and intuitive control of these devices. Such models have often been applied to estimate joint kinematic variables like angles, angular velocities, and accelerations, with the former receiving considerably more attention in the literature. Despite this prevailing focus, we proposed that employing velocities or accelerations as the target variable of myoelectric regressors instead of angles could lead to better regression performances. To investigate this, our study compares the performance of electromyogram (EMG) regressors in predicting each of these three joint kinematic variables. A comprehensive comparison was conducted, including different regression models, feature extraction methods, conditions, joints, and performance metrics. Overall, the performances of models estimating joint angular velocities or accelerations were consistently superior to those estimating angles. However, the differences between angular velocities and accelerations often were small and lacked statistical significance. These results were observed for most models, features, and conditions, not just the best-performing ones (which were: models = convolutional neural networks; features = waveform length, difference absolute mean value, and Willison amplitude; and conditions = wider and faster movements). Several factors were proposed to explain these findings in the context of current literature, including aspects of the EMG-kinematic signals relationships, kinematic waveform properties, studied movements, and the process for obtaining kinematic variables. Based on the results and discussion, recommendations for future works were made, and areas for further research were suggested, both aiming at improving myoelectric assistive technologies.

1. Introduction

To restore mobility of individuals with lost motor function, extensive research has been conducted on assistive devices such as powered prostheses (Farina et al., 2014; Igual, Pardo, Hahne, & Igual, 2019) and exoskeletons (du Plessis, Djouani, & Oosthuizen, 2021). Multiple strategies have been proposed to control these devices. Examples include the integration of vision systems on the palm of prosthetic hands to assist their users in grasping objects (Odeyemi, Ogbeyemi, Wong, & Zhang, 2024), and the use of bioelectric signals recorded from the users of powered prostheses to naturally move such devices. Within

this context, bioelectric signal-based control systems, especially those employing myoelectric control, have gained prominence.

Myoelectric control consists of employing EMG data, which are electrical signals generated during muscle contractions, to control devices such as powered prostheses or exoskeletons. It works by leveraging EMG information to infer a person's motor intention, which can then be used to operate the said devices. Due to its numerous applications for human-machine interaction, particularly in the development of assistive and rehabilitation devices, myoelectric control has the potential

* Corresponding author.

E-mail addresses: eric.becman@usp.br (E.C. Becman), driemeie@usp.br (L. Driemeier), oron.levin@kuleuven.be (O. Levin), stephan.swinnen@kuleuven.be (S.P. Swinnen), aforner@usp.br (A. Forner-Cordero).

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to considerably improve the quality of life for individuals with motor disabilities.

Over the past few decades, pattern recognition myoelectric control has been widely studied, with a focus on machine learning classification models (Bao, Xie, Yang, Zhou, & Zhang, 2022; Fu, Choudhury, Hosseini, Simpson, & Park, 2022). These models are used to establish a relationship between the EMG signal and a set of discrete motion classes, such as in hand gesture recognition problems. However, pattern recognition control differs from the biological control of movements since natural joint motions are continuous and with values that are proportional to the contraction levels of the muscles related to that joint (Farina et al., 2014; Jiang, Dosen, Müller, & Farina, 2012). Such a difference may result in a less intuitive and natural device control for its users. This inherent limitation of pattern recognition has led researchers to explore other control schemes in order to offer proportional myoelectric control.

Regression machine learning models have been receiving growing attention from the scientific community to provide this natural and proportional control (Jiang et al., 2023). These models work by determining a mapping between the EMG signal and a continuous motor-related output, such as joint angles (Lu, Wang, Zhou, Wei, & Xu, 2022). Examples of regression models applied in myoelectric control include decision trees (DTs), support vector machines (SVMs), and multilayer perceptrons (MLPs) (Gu, Lin, Zhang, & Zhao, 2022). Traditionally, such models have been used within the feature engineering framework, where they infer motor intention based on manually computed EMG features as input data (e.g., Pan, Zhang, Liu, Sheng, & Zhu, 2014). More recently, deep learning models operating under the feature learning framework have emerged in the field (Bao et al., 2022; Phinyomark & Scheme, 2018). This approach allows the models to learn how to extract relevant features directly from the EMG data. Among the deep learning models, convolutional neural networks (CNNs) have been widely studied for myoelectric control (e.g., Becman, Driemeier, Levin, Swinnen, & Forner-Cordero, 2023). Recurrent neural networks, like the long short-term memory neural networks (LSTMs), have also gained attention in the literature (e.g., Erazo & Ko, 2023). In addition, other deep neural network architectures have been explored, such as networks combining elements of both CNNs and LSTMs (e.g., Lu et al., 2022), as well as Transformer-inspired networks (e.g., Jiang, Song, Zhang, Song, & Yan, 2022).

A common use for myoelectric regression models is the estimation of joint kinematic variables, which includes joint angles, angular velocities, and angular accelerations. Among these three types of kinematic variables, joint angle estimation seems to be the most studied by a large margin. This observation is supported by a brief literature review reported in Appendix A.

However, despite the prevalence of joint angle estimation in the current literature, it is important to consider the possibility of myoelectric signals being more closely related to other kinematic variables than to joint angles. EMG signals are the sum of the electrical activity of muscle fibers, as detected by the recording system. This activity triggers biochemical changes within the muscle fibers, inducing them to contract. The generated tension forces are transmitted to the bones via connective tissues and tendons, resulting in net torque at a joint spanned by the contracting muscles. Based on Newton's second law for rotation, this net torque is directly proportional to the joint's angular acceleration. When integrated over time, joint acceleration yields joint velocity and, subsequently, joint angles. Therefore, the combined EMG signals from all the muscles acting on a joint could be more directly related to the joint's angular accelerations or velocities than to its angles.

If a joint kinematic variable is more closely related to the EMG signals, it may be easier for regression models to learn the relationship between the EMG and such a variable. In practical terms, this could lead to improved myoelectric regression performance, resulting in a more accurate control scheme. For myoelectric assistive devices, this could

enhance their usability and ultimately improve the quality of life for their users. Then, given the large predominance of joint angle estimation studies, it is important to investigate how myoelectric regression performance might differ if models were employed to estimate other types of joint kinematic data instead.

Thus, in this study, our goal was to investigate whether the performance of EMG regressors varies when predicting different types of kinematic variables. More specifically, we compared the performance of various regressors when estimating joint angles, joint angular velocities, or joint angular accelerations based on EMG data. To accomplish this, we analyzed EMG and motion data from volunteers performing a series of dynamic and cyclic arm motions. The dataset contained recordings of motions performed using two joints and under multiple conditions. Several regression models were examined, including models fed with engineered features and deep learning regressors under the feature learning paradigm. For the feature engineering approach, a wide range of features was evaluated. By considering several types of regression models, features, conditions, and joints, this study provides a comprehensive view of the performances of EMG-based regression models for different types of joint kinematic predictions.

2. Methods

2.1. Data

The data employed in this study is the same used for Becman et al. (2023), which was previously collected by Levin, Forner-Cordero, Li, Ouamer, and Swinnen (2008) with the approval of the Committee for Medical Ethics of the University Hospitals KU Leuven (ML3273).

The dataset consists of synchronized upper-limb motion data and EMG recordings for six healthy volunteers (4 right-handed and 2 left-handed; mean $\pm$ standard deviation: age 28 ± 5 years, height 1.79 ± 0.06 m, mass 74.67 ± 12.99 kg). These measurements were taken as the participants used their dominant arms to draw a star-like shape on a desk in front of them while sitting comfortably on a chair. This star was composed of four intersecting lines with varying orientations (Fig. 1 A). The subjects were instructed to trace these lines using a low-friction pen. For each of the four orientations, the respective line was traced five times. Each of these five repetitions involved forward and backward movements that started and ended at the center of the star (Fig. 1 B). After finishing the five repetitions of one orientation, participants proceeded to the next orientation, following a clockwise (for right-handed volunteers) or counter-clockwise (for left-handed volunteers) order (Figs. 1 C and 1 D). After completing one pass through all orientations, subjects returned to trace each line again using the same five-repetition pattern. This entire cycle of visiting all orientations twice was called a trial (Fig. 1 D). The movements of a trial were performed continuously, with no pauses between or within orientations. All volunteers were asked to complete multiple trials using different conditions. For this, two motion amplitudes were considered, 8 and 16 cm, representing the distance between opposite edges of the star. Similarly, two motion frequencies were considered, 1.33 or 2 Hz, in which one cycle corresponds to tracing one line once (one repetition; Fig. 1 B). A metronome was used to guide the individuals to maintain their motions under the desired frequency. For each of the four possible conditions (combinations of motion amplitude and frequency), four trials were recorded, leading to a total of 16 trials per subject (2 frequencies $\times$ 2 amplitudes $\times$ 4 trials). During the whole data collection, subjects were stabilized to restrict trunk and wrist movements, allowing only shoulder and elbow motions.

Upper-limb movement data were gathered using an Optotrak three-dimensional position tracking system (Northern Digital; Waterloo, Ontario, Canada), which monitored the motion of infrared light-emitting diodes (IREDs) at a 200 Hz sampling rate. As shown in Fig. 1 A, eight IREDs were placed on the participants: two on their shoulders, two on their upper arms, two on their forearms, and two on their

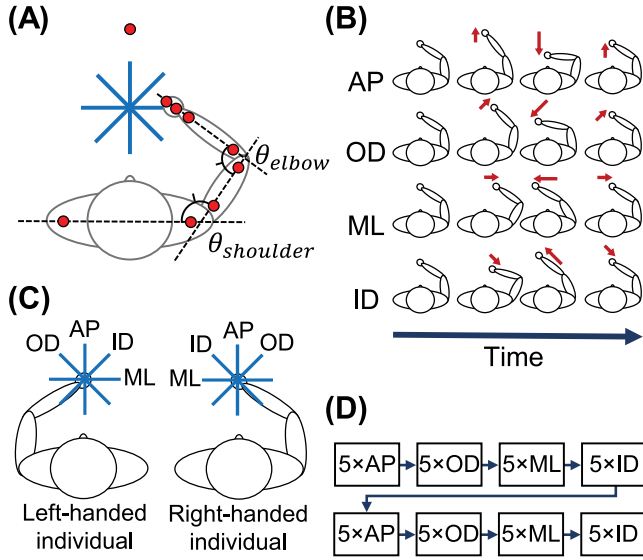


Fig. 1. Procedure for data collection. (A) Diagram illustrating the star-like shape in blue, the arrangement of the infrared light-emitting diodes in red, and the angles for the shoulder and elbow joints ($\theta_{shoulder}$ and θ_{elbow}) for a right-handed subject. (B) Timeline of a single repetition from each of the 4 orientations for a right-handed subject. (C) The line orientations for both left- and right-handed volunteers. (D) The order of the orientations used in a single trial. The four orientations were named anteroposterior (AP), outer-diagonal (OD), mediolateral (ML), and inner-diagonal (ID). Source: Figure reproduced, with permission, from Becman et al. (2023). © 2023 IEEE.

hands. An additional external IRED served as a reference point for the tracking system. The collected motion data were processed using a 15 Hz low-pass second-order Butterworth filter (recursive digital filter).

Surface EMG data from six muscles related to the arm movements were obtained using a Myosystem 2000 recording system (Noraxon; Scottsdale, Arizona, United States). These muscles were the pectoralis, anterior deltoid, posterior deltoid, biceps brachii, triceps brachii, and brachioradialis. EMG bipolar electrodes were placed over the target muscles following the surface electromyography for the non-invasive assessment of muscles (SENIAM) recommendations (Hermens, Freriks, Disselhorst-Klug, & Rau, 2000). Prior to sampling at 1000 Hz, the pre-amplified signal, with a gain of 80 dB, was band-pass filtered between 15 and 500 Hz.

2.2. Joint kinematics

First, the shoulder and elbow joint angles were determined using the upper-body position data from the IRED sensors as shown in Fig. 1 A. Next, joint angular velocities were computed using numerical approximations of the first derivatives of joint angles. Likewise, joint angular accelerations were also estimated as the first derivatives of joint velocities. Finally, given that the EMG and motion data were acquired at different sampling rates, the kinematic signals were linearly interpolated to match the number of samples of the EMGs.

Considering all trials and volunteers, the shoulder joint exhibited minimum and maximum angles of 93.0° and 176.3° , respectively, while the elbow joint showed values ranging from 46.3° to 142.9° . The velocities for the shoulder and elbow joints varied from $-7.5^\circ \text{ s}^{-1}$ to 7.4° s^{-1} and $-10.5^\circ \text{ s}^{-1}$ to $13.9^\circ \text{ s}^{-1}$, respectively. Angular accelerations varied from $-4.1^\circ \text{ s}^{-2}$ to 2.6° s^{-2} for the shoulder and $-4.6^\circ \text{ s}^{-2}$ to 4.5° s^{-2} for the elbow.

2.3. Data segmentation

All EMG signals were segmented into moving windows of 150 ms with overlaps of 100 ms. This window size is within a proposed optimal

range for myoelectric control (Smith, Hargrove, Lock, & Kuiken, 2011) and is commonly used in the literature (Al-Timemy, Khushaba, Bugmann, & Escudero, 2016; Asogbon et al., 2020; Becman et al., 2023; Campbell, Phinyomark, & Scheme, 2020; Tkach, Huang, & Kuiken, 2010). Subsequently, for each of the three kinematic data types (angles, velocities, and accelerations) and the two joints (shoulder and elbow), a single kinematic value was assigned to every EMG window. These individual values were calculated as the averages of the respective joint kinematic variables over the duration of the corresponding window, thereby smoothing out transient kinematic changes within those intervals. Ultimately, this segmentation generated multiple tuples per trial. To be more specific, each tuple included: one EMG window (a 6×150 matrix, corresponding to 150 samples from 6 channels), one value for the shoulder joint angle, one for the elbow joint angle, one for the shoulder joint velocity, one for the elbow joint velocity, one for the shoulder joint acceleration, and one for the elbow joint acceleration.

2.4. Feature engineering

Several feature extraction methods were employed to obtain information from EMG windows. We examined in this article an extensive list of feature extractors drawn or adapted from the literature. A total of 53 extraction methods were considered, which are presented in Table 1. These methods produce features from time-domain, frequency-domain, and non-linear analyses, capturing a variety of EMG properties.

For most of the considered methods, the feature extraction yields only one feature vector per EMG window. Specifically, for each of these extractors, a single feature value was derived separately for each of the six EMG channels of a window, resulting in a 6×1 feature vector. However, other extractors used in this study can produce more than one value per channel. For example, the AR of order 2 produces two values per channel. In such cases, the multiple values from these extractors were analyzed as separate features. For instance, the first coefficient of the AR of order 2 is treated as one feature, while the second coefficient is treated as another feature. Therefore, individual features would still be represented by a 6×1 vector per EMG window. This approach ensured that all feature vectors had the same size, allowing for a fair comparison of their results. The number of individual features that each extractor can produce is also presented in Table 1.

Throughout the rest of the article, an individual feature obtained with a feature extraction method that produces multiple values per EMG channel is referred to using the following terminology: “feature extractor acronym” [“index of the feature within the context of the extractor”/“total number of features produced by the extractor”]. Using again the example of AR of order 2, the first coefficient would be referred to as AR [1/2], and the second coefficient would be AR [2/2].

As reported in Table 1, some of the explored feature extraction methods have hyperparameters (e.g., APEN). The optimization process for selecting the best hyperparameter values for each extractor is described later in Section 2.9.

2.5. Data partitioning

The dataset was divided into distinct subsets for training and performance evaluation of the studied regression models (presented next in Sections 2.7 and 2.8). Specifically, data from trials 1, 2, and 3 were allocated for model training, whereas data from trial 4 were reserved exclusively for testing. For hyperparameter optimization, a separate partitioning scheme was employed: data from trials 1 and 2 served as the training set, and data from trial 3 functioned as the validation set. More details regarding the data partitioning for hyperparameter optimization, model training, and model testing are reported later in Sections 2.9, 2.10, and 2.11, respectively.

Table 1

List of feature extraction methods. This table reports the names and acronyms for all the studied feature extractors, alongside reference examples in which their definitions can be found. Additionally, for every method, this table presents the hyperparameters that had their values selected through hyperparameter optimization (see Section 2.9). For each of these hyperparameters, their names, search ranges used during optimization, and best values found are provided. Finally, the number of individual features produced per method is also included.

Feature extractor names and acronyms	Optimized hyperparameters			Number of resulting features	References
	Names and symbols	Search ranges	Best values		
Approximate entropy (APEN)	• Embedding dimension (m) • Tolerance (r)	• $m \in [1..4]$ • $r \in \{0.05 \cdot x \mid x \in [1..10]\}$	• $m = 1$ • $r = 0.5$	1	Pincus (1991)
Autoregressive coefficients (AR)	• Order (p)	• $p \in [2..6]$	• $p = 2$	p	Zardoshti-Kermani, Wheeler, Badie, and Hashemi (1995)
Cepstral coefficients (CEPS)	• Order (p)	• $p \in [2..6]$	• $p = 2$	p	Phinyomark, Phukpattaranont, and Limsakul (2012)
Coefficient of variation (CV)	—	—	—	1	Adapted from Khushaba, Al-Timemy, Al-Ani, and Al-Jumaily (2017) (see Appendix B)
Difference absolute mean value (DAMV)	—	—	—	1	Abbaspour, Linden, Gholamhosseini, Naber, and Ortiz-Catalan (2020)
Difference absolute standard deviation value (DASDV)	—	—	—	1	Li et al. (2021)
Difference autoregressive coefficients (DAR)	• Order (p)	• $p \in [2..6]$	• $p = 2$	p	Phinyomark et al. (2014)
Difference cepstral coefficients (DCEPS)	• Order (p)	• $p \in [2..6]$	• $p = 2$	p	Phinyomark et al. (2014)
Difference log detector (DLD)	—	—	—	1	Phinyomark et al. (2014)
Difference v-order (DVORD)	• Order (p)	• $p \in [3..6]$	• $p = 3$	1	Phinyomark et al. (2014)
Difference variance (DVAR)	—	—	—	1	Phinyomark et al. (2014)
Higuchi fractal dimension (HFD)	• Maximum value of the parameter k (k_{max})	• $k_{max} \in [2..10]$	• $k_{max} = 6$	1	Arjunan and Kumar (2010)
Histogram frequencies (HIST)	• Number of bins (n)	• $n \in \{3, 5, 7, 9\}$	• $n = 5$	n	Zardoshti-Kermani et al. (1995)
Integrated electromyogram (IEMG)	—	—	—	1	Phinyomark et al. (2012)
Integrated Teager–Kaiser energy operator (ITKEO)	—	—	—	1	Adapted from Khushaba et al. (2017) (see Appendix B)
Irregularity factor (IF)	—	—	—	1	Al-Timemy et al. (2016)
Katz fractal dimension (KFD)	—	—	—	1	Gupta, Suryanarayanan, and Reddy (1997)
Kurtosis (KURT)	—	—	—	1	Abbaspour et al. (2020)
Log detector (LD)	—	—	—	1	Phinyomark et al. (2012)
Maximum fractal length (MFL)	—	—	—	1	Abbaspour et al. (2020)
Mean absolute value (MAV)	—	—	—	1	Arjunan and Kumar (2010)
Mean absolute value slope (MAVS)	—	—	—	1	Phinyomark et al. (2012)
Mean of peaks (MPK)	—	—	—	1	Abbaspour et al. (2020)
Mean power (MNP)	—	—	—	1	Phinyomark et al. (2012)
Mean power frequency (MNF)	—	—	—	1	Phinyomark et al. (2012)
Median power frequency (MDF)	—	—	—	1	Phinyomark et al. (2012)
Modified mean absolute value type 1 (MMAV1)	—	—	—	1	Phinyomark et al. (2012)
Modified mean absolute value type 2 (MMAV2)	—	—	—	1	Phinyomark et al. (2012)
Modified mean absolute value type 3 (MMAV3)	—	—	—	1	Adapted from Phinyomark et al. (2012) (see Appendix B)
Multiple Hamming windows (MHW)	• Number of windows (n) • Overlap percentage (o)	• $n \in \{3, 5, 7, 9\}$ • $o \in \{10 \cdot x \mid x \in [1..9]\}$	• $n = 3$ • $o = 10$	n	Phinyomark et al. (2012)
Myopulse percentage rate (MYOP)	• Threshold (t)	• $t \in [10^{-4}, 10^1]$	• $t = 0.704$	1	Phinyomark et al. (2012)
Number of peaks (NPK)	—	—	—	1	Abbaspour et al. (2020)
Peak power frequency (PKF)	—	—	—	1	Abbaspour et al. (2020)

(continued on next page)

Table 1 (continued).

Feature extractor names and acronyms	Optimized hyperparameters			Number of resulting features	References
	Names and symbols	Search ranges	Best values		
Root mean square (RMS)	—	—	—	1	Li et al. (2021)
Root squared zeroth order moment (RSM0)	—	—	—	1	Al-Timemy et al. (2016)
Root squared second order moment (RSM2)	—	—	—	1	Al-Timemy et al. (2016)
Root squared fourth order moment (RSM4)	—	—	—	1	Al-Timemy et al. (2016)
Sample entropy (SAMPEN)	• Embedding dimension (m) • Tolerance (r)	• $m \in [1..4]$ • $r \in \{0.05 \cdot x \mid x \in [1..10]\}$	• $m = 1$ • $r = 0.5$	1	Richman and Moorman (2000)
Simple square integral (SSI)	—	—	—	1	Li et al. (2021)
Slope sign changes (SSC)	• Threshold (t)	• $t \in \{0\} \cup [10^{-4}, 10^1]$	• $t = 0.077$	1	Abbaspour et al. (2020)
Skewness (SKEW)	—	—	—	1	Abbaspour et al. (2020)
Sparseness (SPARSE)	—	—	—	1	Al-Timemy et al. (2016)
Standard deviation (SD)	—	—	—	1	Abbaspour et al. (2020)
Time-dependent power spectrum descriptors (TDPDSD)	—	—	—	6	Al-Timemy et al. (2016)
Time-domain descriptors (TDD)	—	—	—	7	Adapted from Khushaba et al. (2017) (see Appendix B)
Total power (TTP)	—	—	—	1	Phinyomark et al. (2012)
V-order (VORD)	• Order (p)	• $p \in [3..6]$	• $p = 3$	1	Tkach et al. (2010)
Variance (VAR)	—	—	—	1	Phinyomark et al. (2014)
Variance of central frequency (VCF)	—	—	—	1	Phinyomark et al. (2012)
Waveform length (WL)	—	—	—	1	Tkach et al. (2010)
Waveform length ratio (WLR)	—	—	—	1	Al-Timemy et al. (2016)
Willison amplitude (WAMP)	• Threshold (t)	• $t \in [10^{-4}, 10^1]$	• $t = 0.415$	1	Tkach et al. (2010)
Zero crossings (ZC)	• Threshold (t)	• $t \in \{0\} \cup [10^{-4}, 10^1]$	• $t = 0.670$	1	Tkach et al. (2010)

2.6. Standardization

To avoid possible scale-related issues when optimizing and evaluating different machine learning models, and to ensure that threshold values used for computing features like MYOP and ZC were meaningful and consistent across different EMG channels, volunteers, and conditions, all EMG, feature, and joint kinematic data were standardized.

The standardization process was applied to the data used in the hyperparameter optimization steps (Section 2.9) and to the data used for model training and testing (Sections 2.10 and 2.11). The standardization method was the same for both applications, differing only in terms of the parameters used. During hyperparameter optimization, for each volunteer and condition, all EMG, feature, and motion data from trials 1, 2, and 3 were standardized based on the means and standard deviations of samples from trials 1 and 2. For model training and testing, the data from trials 1, 2, 3, and 4 were standardized based on the means and standard deviations derived from trials 1, 2, and 3. The standardization was performed individually for each EMG channel, feature, joint, and kinematic type. The standardized features were computed from standardized EMG data.

2.7. Traditional machine learning models

In this article, we explore various machine learning frameworks for myoelectric regression, including models that rely on feature engineering. To differentiate these models from the more recent deep learning models under the feature learning scheme, we will refer to the former as traditional models throughout this text. A list of studied traditional models is presented in Table 2, all of which have been previously used in myoelectric control studies (examples are given in the same table).

These traditional models were used to predict joint kinematics based on features manually calculated from standardized EMG windows. For each model type, feature, joint, kinematic type, and EMG window, the respective 6×1 vector of standardized features would be fed into the regression model for estimating the corresponding joint kinematic standardized value.

All of these models required hyperparameter choices. The optimization process for selecting the best hyperparameter configuration for each model is presented in Section 2.9.

All reported traditional models except the MLPs were built, trained, and tested using the Scikit-learn 1.2.1 Python package. MLPs were implemented and used with the TensorFlow 2.10.1 package. The training and testing procedures for these models are explained in Section 2.10 and in Section 2.11, respectively.

2.8. Deep learning machine learning models

Alongside traditional models, we also investigated deep learning models for kinematic regression. Table 3 reports all the studied deep learning models, which have already been used in myoelectric control research, as shown by the examples in the same table.

The selected deep learning models worked under the feature learning framework, which allows the models to learn to identify relevant features directly from raw data. This approach eliminates the need for the model designer to manually determine and compute relevant features from the data. Consequently, the chosen models were used to predict joint kinematic values directly from EMG windows. For each model type, joint, kinematic type, and EMG window, the respective 6×150 matrix of standardized EMG signals was used as the model's

Table 2

List of traditional machine learning models. This table reports the names and acronyms for all the studied traditional models, alongside reference examples in which they were applied for myoelectric control. Additionally, for every model, this table presents the hyperparameters that had their values selected through hyperparameter optimization (see Section 2.9). For each of these hyperparameters, their names, search ranges used during optimization, and best values found are provided.

Model names and acronyms	Optimized hyperparameters			References
	Names and symbols	Search ranges	Best values	
Decision tree (DT)	- Maximum depth of the tree (d) - Minimum number of samples needed to split an internal node (s_s) - Minimum number of samples required to be at a leaf node (s_ℓ) - Maximum number of leaf nodes (ℓ_{max}) - Minimum impurity decrease to split a node (Δ) - Complexity parameter (α)	$d \in \{\infty\} \cup [1..30]$ $s_s \in [2..30]$ $s_\ell \in [1..30]$ $\ell_{max} \in \{\infty\} \cup [1..30]$ $\Delta \in \{0\} \cup [10^{-4}, 10^0]$ $\alpha \in \{0\} \cup [10^{-4}, 10^0]$	$d = 14$ $s_s = 19$ $s_\ell = 13$ $\ell_{max} = 30$ $\Delta = 1.184 \cdot 10^{-2}$ $\alpha = 1.62 \cdot 10^{-4}$	Gu et al. (2022)
Random forest (RF)	- Number of trees (n) - Maximum depth of the tree (d) - Minimum number of samples needed to split an internal node (s_s) - Minimum number of samples required to be at a leaf node (s_ℓ) - Maximum number of leaf nodes (ℓ_{max}) - Minimum impurity decrease to split a node (Δ) - Complexity parameter (α)	$n \in \{50, 75, 125, 150\}$ $d \in \{\infty\} \cup [1..30]$ $s_s \in [2..30]$ $s_\ell \in [1..30]$ $\ell_{max} \in \{\infty\} \cup [1..30]$ $\Delta \in \{0\} \cup [10^{-4}, 10^0]$ $\alpha \in \{0\} \cup [10^{-4}, 10^0]$	$n = 125$ $d = \infty$ $s_s = 30$ $s_\ell = 14$ $\ell_{max} = \infty$ $\Delta = 0$ $\alpha = 0$	Mokri, Bamdad, and Abolghasemi (2022)
K-nearest neighbors (KNN)	- Number of neighbors (k) - Weight function (f)	$k \in [3..50]$ $f \in \{\text{uniform, distance}\}$	$k = 34$ $f = \text{uniform}$	Gu et al. (2022)
Support vector machine (SVM)	- Radial basis function kernel coefficient (γ) - Regularization (c)	$\gamma \in [10^{-4}, 10^4]$ $c \in [10^{-4}, 10^4]$	$\gamma = 1.960 \cdot 10^{-1}$ $c = 1.222 \cdot 10^{-1}$	Lu et al. (2022)
Linear regression (LR)	L2 regularization (λ)	$\lambda \in \{0\} \cup [10^{-4}, 10^4]$	$\lambda = 77.543$	Gu et al. (2022)
Polynomial regression (PR)	- Degree (d) - L2 regularization (λ)	$d \in \{2, 3, 4\}$ $\lambda \in \{0\} \cup [10^{-4}, 10^4]$	$d = 2$ $\lambda = 576.875$	Dai and Hu (2020)
Multilayer perceptron (MLP)	- Number of hidden fully-connected layers (n_d) - Number of neurons per hidden fully-connected layer (n_u) - L2 regularization (λ) - Learning rate (η) - Maximum number of training epochs (t_{max}) - Batch size used for training (b) - Minimum loss improvement for early stopping criterion (ΔL) - Patience for early stopping criterion (Δt)	$n_d \in [1..5]$ $n_u \in \{2^x \mid x \in [5..8]\}$ $\lambda \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ $\eta \in [10^{-5}, 10^0]$ $t_{max} \in \{100, 500, 1000\}$ $b \in \{2^x \mid x \in [5..8]\}$ $\Delta L \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ $\Delta t \in \{10, 15, 20, 25\}$	$n_d = 1$ $n_u = 32$ $\lambda = 3.981 \cdot 10^{-2}$ $\eta = 2.51 \cdot 10^{-4}$ $t_{max} = 1000$ $b = 32$ $\Delta L = 10^{-2}$ $\Delta t = 20$	Lu et al. (2022)

input for estimating the corresponding joint kinematic standardized value.

Similar to traditional models, deep learning models also required hyperparameter selection. The hyperparameter optimization procedure is detailed in Section 2.9.

All the deep learning models were built, trained, and tested using the TensorFlow 2.10.1 Python package. The training and testing procedures for these models are described in Section 2.10 and in Section 2.11, respectively.

While our study focuses on models commonly used in myoelectric control, it should be noted that there are approaches from other disciplines – such as architectures combining convolutional and transformer layers (Wang, Qian, Zhang, Zhang, & Luo, 2023) – that have not been as frequently studied for EMG regression and could provide valuable insights for future research.

2.9. Hyperparameter optimization

In this study, we systematically optimized hyperparameters for both features and models.

2.9.1. Traditional models with feature engineering

The optimization process for traditional machine learning regressors with feature engineering involved alternating rounds of feature and model hyperparameter optimization. This process consisted of four consecutive rounds, with the results of each round being used in the next

one. The rationale here was to select the hyperparameter configuration for each model that would result in the best average performance among the different feature extractors, and vice versa.

Round 1 - Optimizing feature extraction hyperparameters using linear models.

In this round, all feature extraction methods with hyperparameters were individually optimized. For each of these extractors, various hyperparameter settings were tried, and their resulting feature vectors were used as input for linear regression (LR) models. These regressors were then evaluated on validation sets, and the hyperparameters that yielded the best validation performance were selected for the respective feature extraction method. The subsequent paragraphs provide a more detailed description of the steps taken in this process.

First, a specific set of hyperparameters was chosen for a given feature extractor. Then, we used the extractor with these hyperparameters to derive feature vectors from the windowed EMG data of trials 1 and 2, performed by a single volunteer under a particular condition. In the cases where the feature extractor produces more than one feature per EMG channel, each of these features was considered separately. Next, we selected joint kinematic data for one joint and one kinematic type. An LR model without regularization was then trained on the feature vectors to predict the selected kinematic data for the selected joint. Once the LR model was trained, it was used to predict the same type of joint kinematic data based on the feature vectors extracted from the EMG of trial 3 of the same volunteer and condition. Prior to model

Table 3

List of deep learning models. This table reports the names and acronyms for all the studied deep learning models, alongside reference examples in which they were applied for myoelectric control. Additionally, for every model, this table presents the hyperparameters that had their values selected through hyperparameter optimization (see Section 2.9). For each of these hyperparameters, their names, search ranges used during optimization, and best values found are provided.

Model names and acronyms	Optimized hyperparameters			References
	Names and symbols	Search ranges	Best values	
Convolutional neural network (CNN)	• Number of convolutional blocks (n_c) • Number of convolutional kernels (n_k) • Number of hidden fully-connected layers (n_d) • Number of neurons per hidden fully-connected layer (n_u) • Convolutional kernel spatial dimension (k_s) • Convolutional kernel temporal dimension (k_t) • Type of pooling layer used in convolutional blocks ($pool$) • Dropout probability (p) • Learning rate (η) • Maximum number of training epochs (t_{max}) • Batch size used for training (b) • Minimum loss improvement for early stopping criterion (ΔL) • Patience for early stopping criterion (Δt)	• $n_c \in [1..6]$ • $n_k \in \{2^x \mid x \in [3..7]\}$ • $n_d \in [0..6]$ • $n_u \in \{2^x \mid x \in [3..7]\}$ • $k_s \in [1..6]$ • $k_t \in [2..15]$ • $pool \in \{\text{average, maximum}\}$ • $p \in \{0.1 \cdot x \mid x \in [1..9]\}$ • $\eta \in [10^{-5}, 10^0]$ • $t_{max} \in \{100, 500, 1000\}$ • $b \in \{2^x \mid x \in [5..8]\}$ • $\Delta L \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ • $\Delta t \in \{10, 15, 20, 25\}$	• $n_c = 4$ • $n_k = 32$ • $n_d = 5$ • $n_u = 32$ • $k_s = 1$ • $k_t = 2$ • $pool = \text{average}$ • $p = 0.7$ • $\eta = 3.751 \cdot 10^{-4}$ • $t_{max} = 100$ • $b = 32$ • $\Delta L = 10^{-5}$ • $\Delta t = 20$	Becman et al. (2023)
Long short-term memory neural network (LSTM)	• Number of recurrent blocks (n_r) • Number of recurrent nodes (n_n) • Number of hidden fully-connected layers (n_d) • Number of neurons per hidden fully-connected layer (n_u) • Dropout probability (p) • Learning rate (η) • Maximum number of training epochs (t_{max}) • Batch size used for training (b) • Minimum loss improvement for early stopping criterion (ΔL) • Patience for early stopping criterion (Δt)	• $n_r \in [1..6]$ • $n_n \in \{2^x \mid x \in [3..7]\}$ • $n_d \in [0..6]$ • $n_u \in \{2^x \mid x \in [3..7]\}$ • $p \in \{0.1 \cdot x \mid x \in [1..9]\}$ • $\eta \in [10^{-5}, 10^0]$ • $t_{max} \in \{100, 500, 1000\}$ • $b \in \{2^x \mid x \in [5..8]\}$ • $\Delta L \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ • $\Delta t \in \{10, 15, 20, 25\}$	• $n_r = 5$ • $n_n = 64$ • $n_d = 2$ • $n_u = 32$ • $p = 0.3$ • $\eta = 1.218 \cdot 10^{-4}$ • $t_{max} = 100$ • $b = 128$ • $\Delta L = 10^{-4}$ • $\Delta t = 10$	Erazo and Ko (2023)
Convolutional long short-term memory neural network (CNN + LSTM)	• Number of convolutional blocks (n_c) • Number of convolutional kernels (n_k) • Number of recurrent blocks (n_r) • Number of recurrent nodes (n_n) • Number of hidden fully-connected layers (n_d) • Number of neurons per hidden fully-connected layer (n_u) • Convolutional kernel spatial dimension (k_s) • Convolutional kernel temporal dimension (k_t) • Type of pooling layer used in convolutional blocks ($pool$) • Dropout probability (p) • Learning rate (η) • Maximum number of training epochs (t_{max}) • Batch size used for training (b) • Minimum loss improvement for early stopping criterion (ΔL) • Patience for early stopping criterion (Δt)	• $n_c \in [1..6]$ • $n_k \in \{2^x \mid x \in [3..7]\}$ • $n_r \in [1..6]$ • $n_n \in \{2^x \mid x \in [3..7]\}$ • $n_d \in [0..6]$ • $n_u \in \{2^x \mid x \in [3..7]\}$ • $k_s \in [1..6]$ • $k_t \in [2..15]$ • $pool \in \{\text{average, maximum}\}$ • $p \in \{0.1 \cdot x \mid x \in [1..9]\}$ • $\eta \in [10^{-5}, 10^0]$ • $t_{max} \in \{100, 500, 1000\}$ • $b \in \{2^x \mid x \in [5..8]\}$ • $\Delta L \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ • $\Delta t \in \{10, 15, 20, 25\}$	• $n_c = 2$ • $n_k = 32$ • $n_r = 3$ • $n_n = 64$ • $n_d = 1$ • $n_u = 32$ • $k_s = 1$ • $k_t = 6$ • $pool = \text{maximum}$ • $p = 0.6$ • $\eta = 1.480 \cdot 10^{-4}$ • $t_{max} = 100$ • $b = 128$ • $\Delta L = 10^{-5}$ • $\Delta t = 20$	Zhang et al. (2022)

(continued on next page)

training and evaluation, the used EMG, feature, and kinematic data were standardized as described in Section 2.6. The mean squared error (MSE) between the target and predicted values was calculated, yielding one MSE value for the specific combination of feature extractor, hyperparameter configuration, feature, volunteer, condition, joint, and kinematic type. This process was replicated for all features produced by the extractor, for all volunteers, conditions, joints, and kinematic types, resulting in an individual MSE value for each unique combination. We then computed the average MSE value across all previously calculated MSE values, serving as a validation performance metric for the specific set of hyperparameters of the particular feature extraction method.

The steps detailed above result in one MSE value per hyperparameter configuration of a given feature extractor. Therefore, to select the best hyperparameter for the said extractor, these steps were repeated for several different hyperparameter configurations, producing one value for each of them. The configuration with the lowest average MSE value was selected for the given feature extractor.

This entire process was repeated for all considered feature extraction methods.

The steps of this first round can be seen as a specific case of the more general Algorithm 1 for feature extraction hyperparameter optimization, which is also used for round 3 of optimization (explained

Table 3 (continued).

Model names and acronyms	Optimized hyperparameters			References
	Names and symbols	Search ranges	Best values	
Transformer neural network (TNN)	• Number of transformer blocks (n_t) • Number of heads in multi-head self-attention layers (n_h) • Number of neurons in the feed-forward layer of transformer blocks (n_f) • Number of hidden fully-connected layers (n_d) • Number of neurons per hidden fully-connected layer (n_u) • Dropout probability (p) • Learning rate (η) • Maximum number of training epochs (t_{max}) • Batch size used for training (b) • Minimum loss improvement for early stopping criterion (ΔL) • Patience for early stopping criterion (Δt)	• $n_t \in [1..6]$ • $n_h \in \{2^x \mid x \in [1..7]\}$ • $n_f \in \{2^x \mid x \in [3..7]\}$ • $n_d \in [0..6]$ • $n_u \in \{2^x \mid x \in [3..7]\}$ • $p \in \{0.1 \cdot x \mid x \in [1..9]\}$ • $\eta \in [10^{-5}, 10^0]$ • $t_{max} \in \{100, 500, 1000\}$ • $b \in \{2^x \mid x \in [5..8]\}$ • $\Delta L \in \{0\} \cup \{10^x \mid x \in [-5..0]\}$ • $\Delta t \in \{10, 15, 20, 25\}$	• $n_t = 3$ • $n_h = 32$ • $n_f = 64$ • $n_d = 1$ • $n_u = 64$ • $p = 0.1$ • $\eta = 2.078 \cdot 10^{-4}$ • $t_{max} = 100$ • $b = 128$ • $\Delta L = 10^{-5}$ • $\Delta t = 10$	Jiang et al. (2022)

Algorithm 1. Hyperparameter optimization for feature extractors.

Input: List of feature extractors; list of hyperparameter configurations for each feature extractor; list of traditional regression models; list of hyperparameter configurations for each traditional regression model; list of volunteers; list of conditions; EMG data; motion data.

Output: Optimal hyperparameters for each feature extractor.

```

1: for each feature extractor do
2:   for each feature extractor hyperparameter configuration do
3:     Configure feature extractor with hyperparameters;
4:     for each feature that the feature extractor can extract do
5:       for each volunteer do
6:         for each condition do
7:           Select corresponding EMG data from trials 1, 2, and 3;
8:           Standardize the selected EMG data based on statistics from trials 1 and 2;
9:           Extract feature vectors from the standardized EMG data;
10:          Standardize the feature vectors based on statistics from trials 1 and 2;
11:          for each joint do
12:            for each kinematic type do
13:              Select corresponding joint kinematic data from trials 1, 2, and 3;
14:              Standardize the selected joint kinematic data based on statistics from trials 1 and 2;
15:              for each model do
16:                for each model hyperparameter configuration do
17:                  Configure model with hyperparameters;
18:                  Train model on the feature vectors and joint kinematic data from trials 1 and 2;
19:                  Predict joint kinematic data using the trained model and feature vectors from trial 3;
20:                  Compute the MSE between target and predicted values;
21:                  Store the MSE for the current feature extractor, feature extractor hyperparameters, feature, volunteer, condition, joint, kinematic type, model, and model hyperparameters;
22:            Compute the average MSE across all stored MSE values for the current feature extractor hyperparameter configuration;
23:          Select the hyperparameter configuration with the lowest average MSE for the current feature extractor;

```

later in the text). In the case of round 1, the list of models of Algorithm 1 is composed only by an LR, which is associated only with one hyperparameter configuration, an L2 regularization equal to 0 (i.e., no regularization).

Round 2 - Optimizing model hyperparameters using all features.

In the second round, traditional machine learning methods were optimized under the same rationale used in round 1. For every model, multiple hyperparameter configurations were tried in conjunction with each individual studied feature. Here, the feature extractors were configured with the hyperparameters found in the last round. These regressors were then evaluated on validation sets (data from trial 3), and the hyperparameters that produced the best validation performance were selected for the respective model. A detailed list of the steps taken in this round to optimize the models' hyperparameters can be found in Algorithm 2. Kinematic, EMG, and feature data were standardized before model training and evaluation, as described in Section 2.6.

Round 3 - Optimizing feature extraction hyperparameters using all models.

Similar to round 1, we optimized hyperparameters for feature extraction methods. The steps undertaken in this round are represented by Algorithm 1. Differently from round 1, the features here were evaluated using all traditional regressors. Each model was configured with their respective hyperparameters obtained in the previous round.

Round 4 - Optimizing model hyperparameters using all features.

This round focused again on optimizing the hyperparameters for the regressors using the same procedure from round 2, described in Algorithm 2. In round 4, however, the feature extraction methods were configured with the hyperparameters found from the last round.

2.9.2. Deep learning models with feature learning

The hyperparameter optimization for deep learning models shares many similarities with that of traditional machine learning models.

Algorithm 2. Hyperparameter optimization for traditional regression models.

Input: List of traditional regression models; list of hyperparameter configurations for each traditional regression model; list of feature extractors; list of hyperparameter configurations for each feature extractor; list of volunteers; list of conditions; EMG data; motion data.

Output: Optimal hyperparameters for each traditional regression model.

```

1: for each model do
2:   for each model hyperparameter configuration do
3:     for each feature extractor do
4:       for each feature extractor hyperparameter configuration do
5:         Configure feature extractor with hyperparameters;
6:         for each feature that the feature extractor can extract do
7:           for each volunteer do
8:             for each condition do
9:               Select corresponding EMG data from trials 1, 2, and 3;
10:              Standardize the selected EMG data based on statistics from trials 1 and 2;
11:              Extract feature vectors from the standardized EMG data;
12:              Standardize the feature vectors based on statistics from trials 1 and 2;
13:              for each joint do
14:                for each kinematic type do
15:                  Select corresponding joint kinematic data from trials 1, 2, and 3;
16:                  Standardize the selected joint kinematic data based on statistics from trials 1 and 2;
17:                  Configure model with hyperparameters;
18:                  Train model on the feature vectors and joint kinematic data from trials 1 and 2;
19:                  Predict joint kinematic data using the trained model and feature vectors from trial 3;
20:                  Compute the MSE between target and predicted values;
21:                  Store the MSE for the current model, model hyperparameters, feature extractor, feature extractor hyperparameters,
                     feature, volunteer, condition, joint, and kinematic type;
22:             Compute the average MSE across all stored MSE values for the current model hyperparameter configuration;
23:             Select the hyperparameter configuration with the lowest average MSE for the current model;

```

However, there were no alternate rounds of feature and model optimization as the deep learning models worked under the feature learning framework, receiving EMG window data as input.

Initially, for each deep learning model, a specific set of hyperparameters was selected. The model was then trained with the chosen hyperparameters separately for each combination of volunteer, condition, joint, and kinematic data type. The training relied on EMG windows and joint kinematic values from trials 1 and 2. To validate the models, data from trial 3 was used, and the MSE between the expected and predicted values was calculated. This process resulted in individual MSE values for each unique combination of volunteer, condition, joint, and kinematic data type. Next, the average MSE value was computed over all the calculated MSE values for the model hyperparameter configuration. This average MSE served as the validation performance metric for the specific configuration. The process was repeated for multiple sets of hyperparameters of the model. The hyperparameter configuration that produced the lowest average MSE value was chosen for the respective model. All deep learning models underwent the same hyperparameter optimization procedure. Kinematic and EMG data standardization was applied before model training and evaluation as described in Section 2.6.

2.9.3. Model hyperparameters' sensitivity to outliers

The choice of hyperparameters plays a central role in determining the performance of a model, influencing its position on the bias-variance spectrum and its ability to generalize. Hyperparameter optimization is essential to avoid models that are either too simple or too complex for a given task and data. However, it is important to recognize the possibility that a particular hyperparameter configuration may result in an extreme MSE value for a given validation set, which could misguide the selection process. Our optimization process addresses this issue by not relying on single-scenario evaluations. Instead, it averages several MSE values across a wide range of scenarios, including multiple features, volunteers, conditions, joints, and kinematic variables. This approach mitigates the disproportionate impact that a single outlier

can have on the overall evaluation, thereby more reliably identifying configurations that can generalize well across different scenarios.

2.9.4. Search methods

The hyperparameters optimized for all feature extractors are reported in Table 1, while those for traditional models are in Table 2, and those for deep learning models are in Table 3. These tables also present the search interval of each hyperparameter during the search for optimal hyperparameters, as well as the best values for each hyperparameter chosen after optimization.

Both grid and random searches were employed depending on whether the optimized hyperparameters were continuous or discrete. When all optimized hyperparameters were discrete, such as for the feature VORD, grid search was used. When continuous hyperparameters were optimized, as for the feature ZC and the model CNN, random search was used. In the cases where the continuous hyperparameters had a search range spanning different orders of magnitude, such as the threshold for MYOP and the learning rate for neural networks, the random search was conducted for them on a logarithmic scale.

2.9.5. Overview of the obtained models and features

As previously mentioned, the non-neural network models (all traditional regressors except MLP) were implemented using the Scikit-learn 1.2.1 Python package. Apart from the hyperparameters chosen for each of these traditional models (Table 2), the remaining arguments for building and using the models were the package's default values.

For the MLP regressor, the selected architecture consisted of 1 hidden fully-connected (FC) layer with 32 neurons and rectified linear unit (ReLU) activation, followed by an output FC layer with a linear activation and a single neuron, responsible for predicting joint kinematic values (Fig. 2 A).

The obtained CNN architecture comprised 4 convolutional blocks followed by a series of FC layers (Fig. 2 B). Each convolutional block contained a convolutional layer with 32 kernels of size 1×2 and ReLU activation, followed by an average pooling layer with an 1×2 kernel.

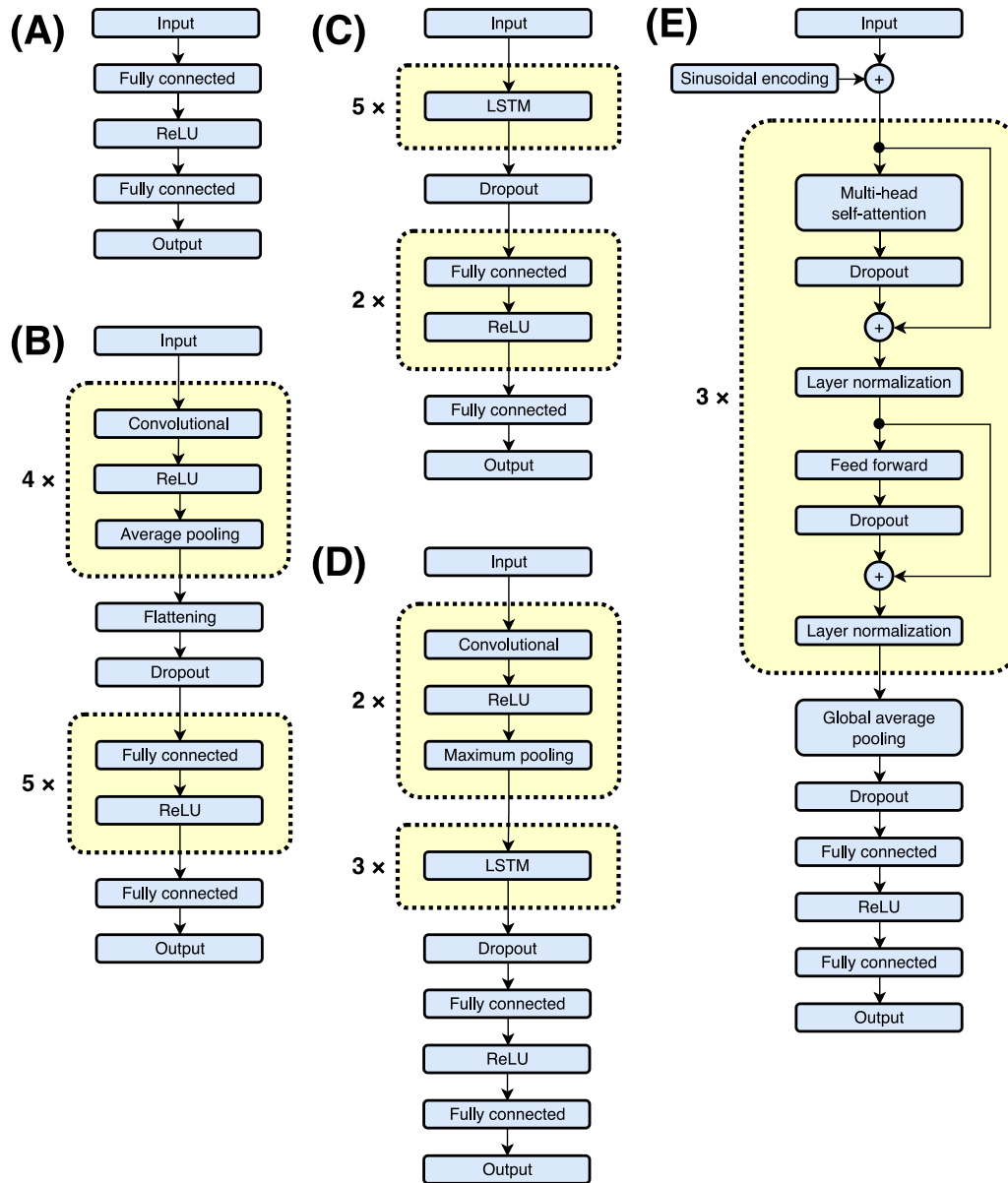


Fig. 2. Diagrams of the selected neural networks. The architecture of MLPs, CNNs, LSTMs, CNN + LSTMs, and TNNs are represented in (A), (B), (C), (D), and (E), respectively. More details on the TNN encoder layers, those within the yellow box in (E), can be found in [Vaswani et al. \(2017\)](#).

The last convolutional block's output array is then flattened and passed to a dropout layer with a 70 % drop probability. Next, the output of the previous layer is fed into a series of 5 hidden FC layers with 32 neurons and ReLU activation. The final layer of the networks consists of an FC layer with a linear activation and 1 neuron, which returns the predicted joint kinematic data.

The first part of the LSTM models included 5 LSTM blocks of 64 nodes each, using the implementation of the TensorFlow 2.10.1 package. The first four blocks received and generated time series of the same length as the EMG windows (150 samples). In the 5th block, only the values produced at the end of the time series were passed to the next layer, which was a dropout layer with a probability of 70%. This layer was then followed by 2 hidden FC layers with 32 neurons and ReLU activation. The output layer is an FC layer with a linear activation similar to those of the other networks (Fig. 2 C).

Convolutional long short-term memory neural network (CNN + LSTM) regressors combined elements from both the CNN and LSTM models (Fig. 2 D). First, the EMG windows were processed by convolutional blocks. Next, the output of the convolutional blocks was

passed to LSTM blocks. The output of the last block was then passed through a dropout layer. Finally, hidden FC layers with ReLU activation were followed by another FC layer with linear activation, which was responsible for outputting joint kinematic values. The hyperparameters chosen for the convolutional long short-term memory neural networks (CNN + LSTMs) are presented in [Table 3](#).

The transformer neural network (TNN) architecture was similar to that originally proposed by [Vaswani et al. \(2017\)](#) (Fig. 2 E). Since their model was applied to a sequence-to-sequence problem (language translation), they used an architecture consisting of both an encoder and a decoder. However, as our study focuses on a sequence-to-one regression task, we employed only the encoder part of their architecture. This encoder produced an output that was then flattened by a global average pooling over the temporal dimension, which was followed by a dropout layer. Subsequently, the outputs of the previous layers were fed into a hidden FC layer with ReLU activation, followed by an FC output layer with linear activation that predicted joint kinematic data. The network used dropout layers for regularization, both after the transformer blocks and within them (as used in [Vaswani et al.](#),

2017), with 10% drop probability. To embed time information into the network inputs, sinusoidal position encoding was added to the EMG windows, also as in Vaswani et al. (2017). The hyperparameters used for this network are presented in Table 3.

For all of the neural networks studied, training was performed using the Adam algorithm, and the hyperparameters learning rate, L2 regularization (for MLPs), batch size, and maximum number of epochs were selected as per Tables 2 and 3. Furthermore, an early stopping criterion was employed to interrupt training in the cases where the MSE loss value did not improve by at least a certain value within a given number of epochs. These two early stopping hyperparameters, the minimum loss improvement value and the number of epochs, were also selected as shown in Tables 2 and 3. Regarding the feature extraction methods, since some of these methods produced multiple features, the total number of features was greater than the number of extractors (53). Given the selected feature extraction hyperparameters (shown in Table 1), these methods combined generated 74 features.

2.10. Model training

Traditional models were trained multiple times, one for each combination of volunteer, condition, joint, kinematic data type, and feature. For each of these combinations, a model was trained to predict standardized joint kinematic values based on standardized feature vectors computed from the standardized EMG data of trials 1, 2, and 3. Altogether, each type of model was trained 10656 times (6 volunteers $\times$ 2 frequencies $\times$ 2 amplitudes $\times$ 2 joints $\times$ 3 kinematic types $\times$ 74 features).

Deep learning models were trained in a similar fashion; however, they did not utilize handcrafted features. Instead, they were trained to predict standardized joint kinematic values from trials 1, 2, and 3 using standardized EMG windows. In total, each model type was trained 144 times (6 volunteers $\times$ 2 frequencies $\times$ 2 amplitudes $\times$ 2 joints $\times$ 3 kinematic types).

Neural network training was performed using the Adam algorithm and an early stopping criterion with the obtained optimal hyperparameter values.

2.11. Model testing

Each of the trained traditional models was tested with data from trial 4 to predict standardized joint kinematic data based on standardized feature vectors from the respective combination of volunteer, condition, joint, kinematic data type, and feature. The same procedure was followed for the deep learning models, but the predicted values were computed instead using standardized EMG windows from the corresponding combination of volunteer, condition, joint, and kinematic data type.

The obtained predicted values were compared with their associated target values using the methodology established by Becman et al. (2023), in which three dimensionless metrics were used to evaluate prediction quality: normalized root mean squared error (NRMSE), Pearson's correlation (CORR), and slope of the linear regression (SLOPE) between target and predicted values. The NRMSE metric was calculated by finding the MSE between target and predicted values, taking the square root of the MSE, and dividing it by the standard deviation (SD) of the target values. It represents the deviation of the predicted values from the target values. This non-negative metric approaches zero as predictions improve. The CORR metric represents the strength of the linear association between the target and predicted values, indicating how well the predicted values follow the trends in the target values. Values of this metric are bounded between -1 and 1 , with 1 corresponding to perfect predictions. The SLOPE metric shows whether the predicted values overestimate (values greater than 1) or underestimate (values less than 1) the variations in the target values, providing insight into the sensitivity of the model to changes in the target values. Ideally, SLOPE should be equal to 1 , indicating that any change in the target values is met with an equal variation in the predicted values.

2.12. Statistics

To investigate whether the performance of EMG regressors varies when they are used to predict different types of joint kinematic variables, we analyzed the performance metrics (NRMSE, CORR, and SLOPE) computed for the predictions obtained for each of the multiple EMG features, regression models, conditions in which the data were collected, and upper-limb joints. Together, these three metrics acted as a proxy for prediction quality. By analyzing them across these multiple scenarios, we could gain a more comprehensive understanding of the relationship between the prediction of different joint kinematic data and myoelectric regression performance. Thus, our results were not tied to only one specific combination of feature, model, condition, and joint.

Statistical analyses were carried out to assess the factors that may have influenced the average values of the performance metrics. For this, we primarily used repeated measures multivariate analysis of variance (MANOVA). However, if we were to conduct the statistical analysis considering all the possible factors (kinematic type, feature, model, condition, and joint) and their interaction effects at the same time, several high-order interaction terms would arise, which could severely compromise the interpretability of the results. Furthermore, many of these terms would be of no interest given the article's objective. Then, for better interpretability and simplicity, the performance differences associated with each kinematic type were analyzed in conjunction with only one other factor at a time. This approach limited the interaction effects to two-way interaction terms, making the analyses more manageable and understandable.

In the cases where statistically significant effects were observed, univariate analyses of variance (ANOVAs) for each individual metric and paired t -tests were used as *post-hoc* analyses. The p -values found from these multiple *post-hoc* comparisons were corrected using the False Discovery Rate adjustment method (Benjamini & Hochberg, 1995). In the context of paired t -tests, the differences between the means of two samples are denoted throughout the text by the symbol $\Delta\bar{X}$.

Data normality assumptions were verified using Shapiro-Wilk tests before performing the previously described analyses, which indicated no meaningful deviations from normality. The significance levels used throughout this study were 5%.

3. Results

3.1. Kinematic types and engineered features

Fig. 3 shows the performance metric values related to every kinematic type and engineered feature. Overall, the performance associated with joint angles was consistently worse than that of joint acceleration and velocities. This gap is more noticeable for NRMSE values, but it is still apparent for the other two metrics. Regarding the features, DAMV, DLD, RSM4, WAMP, and WL were among the best five features for all three metrics. On the other hand, CV, HIST [1/5], HIST [5/5], SKEW, and TDD [6/7] were among the worst five features, displaying poor results across all metrics.

The average prediction quality of traditional models was observed to be dependent on the used input features, the type of output kinematic data, and the interaction between both effects (feature: Pillai's trace = 1.997, $F = 9.958$, $p < 0.001$; kinematic type: Pillai's trace = 1.201, $F = 4.509$, $p = 0.006$; feature $\times$ kinematic type: Pillai's trace = 1.037, $F = 2.642$, $p < 0.001$). This interaction effect was further supported by conducting an ANOVA for each metric individually (NRMSE: $F = 3.210$, $p < 0.001$; CORR: $F = 2.057$, $p < 0.001$; SLOPE: $F = 2.288$, $p < 0.001$).

Consequently, paired t -tests were performed to compare the metric values between different kinematic types given each separate feature. The results are summarized in Table 4. For all considered features and metrics, models that predicted joint accelerations performed better on average than those that estimated joint angles, with most comparisons being statistically significant. Likewise, joint velocity estimation led to

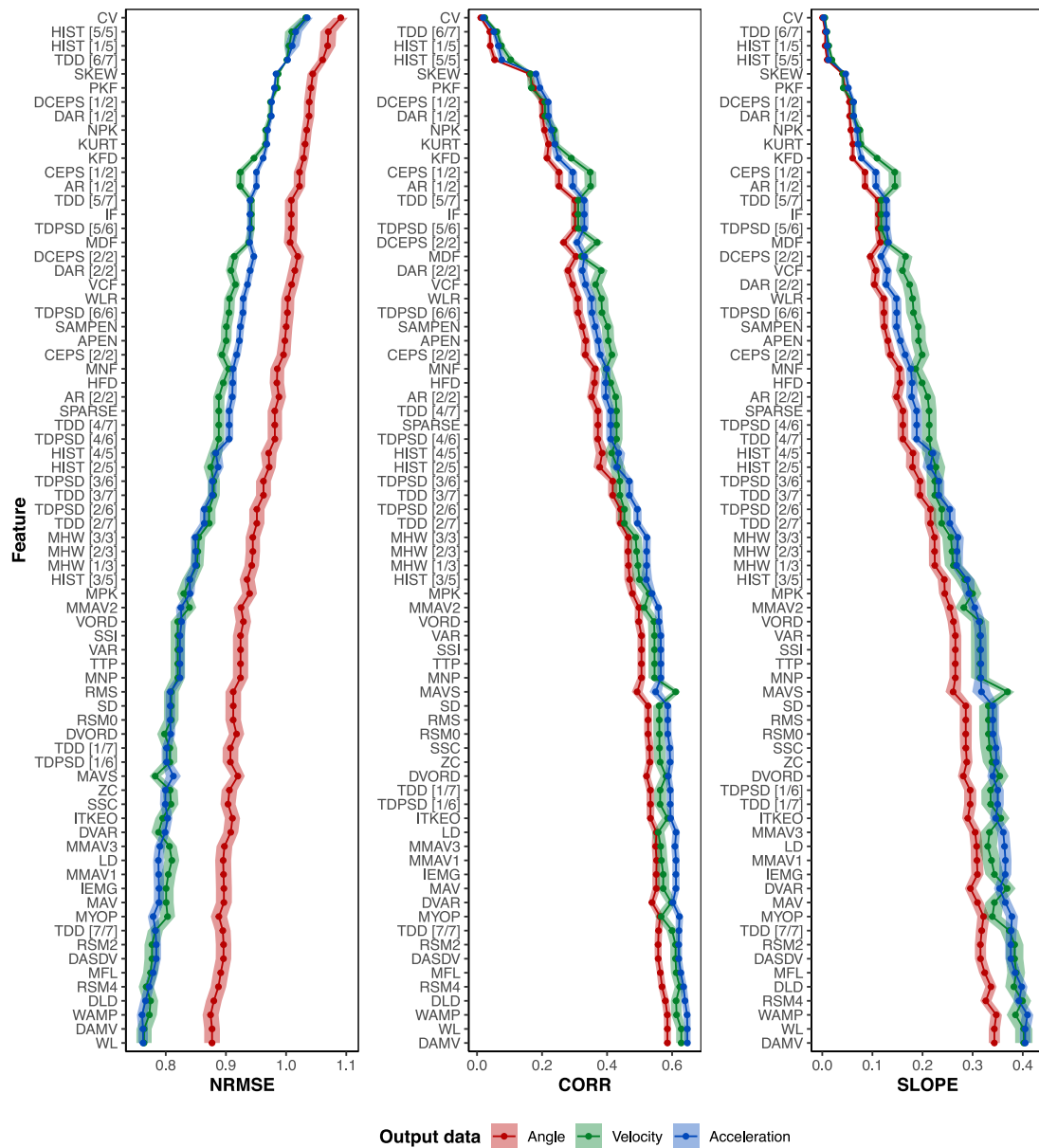


Fig. 3. Performance metric values presented by EMG feature and type of kinematic output data. The points and lines represent the averages computed across the values obtained from all volunteers, conditions, joints, and traditional machine learning models. The shaded areas correspond to the 95% confidence intervals of these averages. To facilitate visualization, features were ordered according to their average metric values. Likewise, feature averages and confidence intervals were shown using continuous lines, despite the variable being categorical, for easier interpretation.

better average values than angle prediction for most feature and metric combinations. However, in this case, only NRMSE had the majority of the tests resulting in significant differences. When comparing joint accelerations and velocities, the former showed better average CORR and SLOPE values for more features, whereas the latter had more features with better NRMSE values. Yet, very few of these comparisons were statistically significant.

Furthermore, since one is often interested in the features that result in best model performances, we also report a summary in Table 5 of the paired *t*-test outcomes only for the top-performing 50% of the features based on their overall values for each metric. The results were very similar to those of Table 4, with the main difference being that joint accelerations showed better average values than joint velocities for all three metrics instead of two. Nonetheless, all of these differences between both kinematic types lacked statistical significance.

3.2. Best engineered feature per traditional model

To facilitate handling both traditional and deep learning models together in subsequent analyses, and since feature selection is often done in the context of feature engineering, we identified and selected the feature that produced the best metric values for each traditional model. Such features are shown in Table 6. For six of seven traditional regressors, the same feature yielded the best values for all three metrics. Consequently, these features were selected as the best features for the respective models. These best features were DAMV for DT and random forest (RF), WAMP for LR and polynomial regression (PR), and WL for k-nearest neighbors (KNN) and MLP. As for the SVM, the feature WL had better results for two of the three metrics (NRMSE and CORR) and was thus chosen as the best feature for the model.

Table 4

Overview of the *post-hoc* paired *t*-test results comparing average regression performance between the three types of output kinematic data, with these tests being conducted independently for each EMG feature and performance metric. This table summarizes the obtained outcomes, presenting the total number of tests and the count of statistically significant differences (based on corrected *p*-values). Comparisons between two kinematic data types are separated into two rows: one row for when one of the kinematic types had better average values and another row for when the other kinematic type had better values. Abbreviations: # = number; signif. = significant.

Comparison		NRMSE		CORR		SLOPE	
Better values	Worse values	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests
Acceleration	Velocity	0	30	0	50	0	40
Velocity	Acceleration	2	44	0	24	0	34
Acceleration	Angle	74	74	46	74	37	74
Angle	Acceleration	–	–	–	–	–	–
Velocity	Angle	71	74	3	73	3	73
Angle	Velocity	–	–	0	1	0	1

Table 5

Overview of the *post-hoc* paired *t*-test results comparing average regression performance between the three types of output kinematic data, with these tests being conducted independently for each EMG feature and performance metric; only results from the top-performing 50% of the EMG features are reported here. This table summarizes the obtained outcomes, presenting the total number of tests and the count of statistically significant differences (based on corrected *p*-values). Comparisons between two kinematic data types are separated into two rows: one row for when one of the kinematic types had better average values and another row for when the other kinematic type had better values. Abbreviations: # = number; signif. = significant.

Comparison		NRMSE		CORR		SLOPE	
Better values	Worse values	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests
Acceleration	Velocity	0	21	0	36	0	27
Velocity	Acceleration	0	16	0	1	0	10
Acceleration	Angle	37	37	37	37	35	37
Angle	Acceleration	–	–	–	–	–	–
Velocity	Angle	37	37	1	37	1	37
Angle	Velocity	–	–	–	–	–	–

Table 6

Best EMG features found for each traditional model and performance metric. The best values were selected considering the average metric values computed over all volunteers, conditions, joints, and kinematic data types.

Model	NRMSE		CORR		SLOPE	
	Best feature	Mean $\pm$ SD	Best feature	Mean $\pm$ SD	Best feature	Mean $\pm$ SD
DT	DAMV	0.845 $\pm$ 0.112	DAMV	0.562 $\pm$ 0.115	DAMV	0.363 $\pm$ 0.116
KNN	WL	0.771 $\pm$ 0.118	WL	0.655 $\pm$ 0.109	WL	0.417 $\pm$ 0.123
LR	WAMP	0.828 $\pm$ 0.118	WAMP	0.579 $\pm$ 0.130	WAMP	0.342 $\pm$ 0.131
MLP	WL	0.795 $\pm$ 0.115	WL	0.629 $\pm$ 0.111	WL	0.374 $\pm$ 0.120
PR	WAMP	0.811 $\pm$ 0.111	WAMP	0.616 $\pm$ 0.113	WAMP	0.319 $\pm$ 0.108
RF	DAMV	0.776 $\pm$ 0.124	DAMV	0.649 $\pm$ 0.113	DAMV	0.441 $\pm$ 0.124
SVM	WL	0.773 $\pm$ 0.122	WL	0.655 $\pm$ 0.107	DAMV	0.444 $\pm$ 0.125

3.3. Kinematic types and models

For each kinematic type, Fig. 4 presents metric values for both traditional models with their selected best features and deep learning models. In general, the ability of models to predict joint angles was worse than the ability to predict joint acceleration and velocities. While this difference is more apparent for NRMSE values, it is still evident in the other two metrics. As to the models' predictive abilities, the CNN had the best average performances among all models according to the three studied metrics. In contrast, the TNN was the deep learning model with the worst performances for all metrics. Between the traditional regressors, the KNN, SVM, and RF models were the ones with the best metric values. Conversely, the DT, LR, and PR models showed the worst average results for the traditional models.

According to the MANOVA, the average metric values were influenced by the regression model used, the type of output kinematic data, and their interaction (model: Pillai's trace = 2.240, $F = 14.741$, $p < 0.001$; kinematic type: Pillai's trace = 1.352, $F = 6.254$, $p = 0.001$; model $\times$ kinematic type: Pillai's trace = 0.781, $F = 1.761$, $p = 0.001$). However, only the two first effects were statistically significant when each metric was analyzed separately using ANOVAs (model – NRMSE: $F = 18.481$, $p < 0.001$; CORR: $F = 43.404$, $p < 0.001$; SLOPE: $F = 111.069$, $p < 0.001$ – kinematic type – NRMSE: $F = 30.122$, $p < 0.001$; CORR:

$F = 5.780$, $p = 0.024$; SLOPE: $F = 5.547$, $p = 0.024$ – model $\times$ kinematic type – NRMSE: $F = 1.572$, $p = 0.225$; CORR: $F = 0.903$, $p = 0.584$; SLOPE: $F = 1.128$, $p = 0.501$). Thus, we proceeded to analyze these two significant effects separately through multiple paired *t*-tests.

Joint accelerations results were found to be better than those of joint angles, with significance obtained for all metrics (NRMSE: $\Delta\bar{X} = -0.121$, $t = 8.509$, $p = 0.001$; CORR: $\Delta\bar{X} = 0.064$, $t = -6.462$, $p = 0.004$; SLOPE: $\Delta\bar{X} = 0.063$, $t = -5.398$, $p = 0.009$). A similar pattern was observed by comparing velocities and angles, with better performances associated with the former. However, significance was only attained for NRMSE (NRMSE: $\Delta\bar{X} = -0.113$, $t = 4.889$, $p = 0.007$; CORR: $\Delta\bar{X} = 0.041$, $t = -1.624$, $p = 0.248$; SLOPE: $\Delta\bar{X} = 0.052$, $t = -2.026$, $p = 0.148$). Moreover, for the three metrics, joint acceleration prediction performances were slightly better on average than those for joint velocities, although these differences were not statistically significant (NRMSE: $\Delta\bar{X} = -0.008$, $t = 0.594$, $p = 0.579$; CORR: $\Delta\bar{X} = 0.023$, $t = -1.222$, $p = 0.276$; SLOPE: $\Delta\bar{X} = 0.011$, $t = -0.528$, $p = 0.62$).

Table 7 shows the results of paired *t*-tests comparing CNNs, models which exhibited better prediction quality, versus the other examined models. The CNNs had significantly better NRMSE values than the PR, LR, LSTM, TNN, and DT models. Regarding CORR, the CNNs significantly outperformed all models except the CNN + LSTM. Finally, the CNNs had significantly better SLOPE values than all other models.

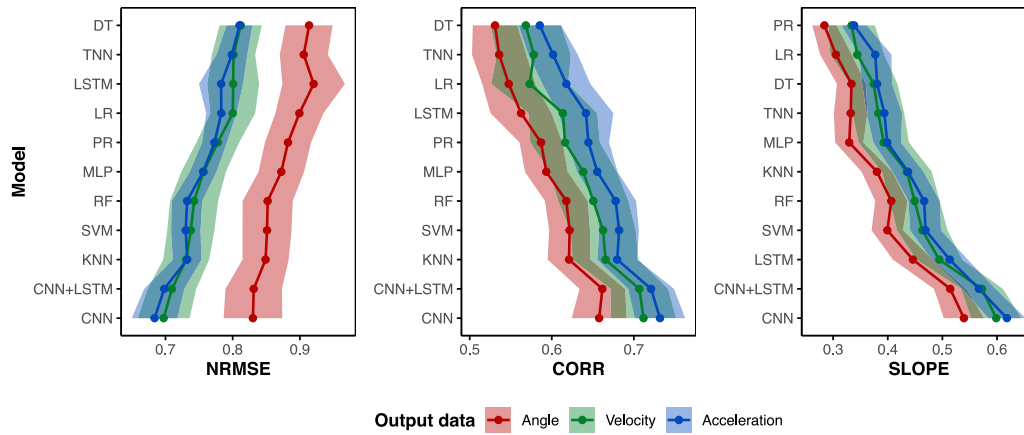


Fig. 4. Performance metric values presented by regression model and type of kinematic output data. In the case of the traditional models, only those trained with their best features are portrayed in the figure. The points and lines represent the averages computed across the values obtained from all volunteers, conditions, and joints. The shaded areas correspond to the 95% confidence intervals of these averages. To facilitate visualization, models were ordered according to their average metric values. Likewise, model averages and confidence intervals were shown using continuous lines, despite the variable being categorical, for easier interpretation.

Table 7

Paired *t*-test results comparing the average prediction performance of CNN *versus* all the other studied models. Results are presented by metrics and models to which the CNNs were compared. For each metric, these compared models are ordered from the best to the worst average metric values computed over all volunteers, conditions, joints, and kinematic data types. The reported *p*-values were corrected for multiple *post-hoc* comparisons. The cells highlighted in yellow correspond to statistically significant differences.

NRMSE		CORR		SLOPE	
Comparison	Test result	Comparison	Test result	Comparison	Test result
CNN + LSTM	• $\Delta\bar{X} = -0.009$ • $t = -1.585$ • $p = 0.174$	CNN + LSTM	• $\Delta\bar{X} = 0.004$ • $t = 0.627$ • $p = 0.558$	CNN + LSTM	• $\Delta\bar{X} = 0.034$ • $t = 2.743$ • $p = 0.041$
KNN	• $\Delta\bar{X} = -0.034$ • $t = 1.770$ • $p = 0.152$	KNN	• $\Delta\bar{X} = 0.045$ • $t = -3.637$ • $p = 0.019$	LSTM	• $\Delta\bar{X} = 0.101$ • $t = 8.882$ • $p < 0.001$
SVM	• $\Delta\bar{X} = -0.036$ • $t = 1.966$ • $p = 0.134$	SVM	• $\Delta\bar{X} = 0.045$ • $t = -3.328$ • $p = 0.023$	SVM	• $\Delta\bar{X} = 0.142$ • $t = -9.386$ • $p < 0.001$
RF	• $\Delta\bar{X} = -0.039$ • $t = 2.094$ • $p = 0.129$	RF	• $\Delta\bar{X} = 0.052$ • $t = -4.010$ • $p = 0.014$	RF	• $\Delta\bar{X} = 0.145$ • $t = -10.066$ • $p < 0.001$
MLP	• $\Delta\bar{X} = -0.058$ • $t = 2.781$ • $p = 0.065$	MLP	• $\Delta\bar{X} = 0.071$ • $t = -4.788$ • $p = 0.008$	KNN	• $\Delta\bar{X} = 0.168$ • $t = -10.909$ • $p < 0.001$
PR	• $\Delta\bar{X} = -0.074$ • $t = 3.520$ • $p = 0.034$	PR	• $\Delta\bar{X} = 0.085$ • $t = -5.951$ • $p = 0.004$	MLP	• $\Delta\bar{X} = 0.212$ • $t = -12.263$ • $p < 0.001$
LR	• $\Delta\bar{X} = -0.090$ • $t = 4.329$ • $p = 0.018$	LSTM	• $\Delta\bar{X} = 0.095$ • $t = 20.147$ • $p < 0.001$	TNN	• $\Delta\bar{X} = 0.216$ • $t = 12.008$ • $p < 0.001$
LSTM	• $\Delta\bar{X} = -0.098$ • $t = -17.795$ • $p < 0.001$	LR	• $\Delta\bar{X} = 0.121$ • $t = -8.447$ • $p = 0.001$	DT	• $\Delta\bar{X} = 0.223$ • $t = -11.276$ • $p < 0.001$
TNN	• $\Delta\bar{X} = -0.098$ • $t = -4.853$ • $p = 0.017$	TNN	• $\Delta\bar{X} = 0.129$ • $t = 8.974$ • $p = 0.001$	LR	• $\Delta\bar{X} = 0.243$ • $t = -12.909$ • $p < 0.001$
DT	• $\Delta\bar{X} = -0.108$ • $t = 4.803$ • $p = 0.017$	DT	• $\Delta\bar{X} = 0.139$ • $t = -8.249$ • $p = 0.001$	PR	• $\Delta\bar{X} = 0.267$ • $t = -12.404$ • $p < 0.001$

3.4. Kinematic types and conditions

The performances of models when estimating joint kinematics are shown for each different condition and metric in Fig. 5. For three of the four conditions, worse performances were seen when predicting joint angles than when estimating the other kinematic types. For the amplitude = 8 cm and frequency = 1.33 Hz condition, average CORR

and SLOPE values were similar between joint angle and velocity predictions. Moreover, according to Fig. 5, better model predictions seem to be associated with conditions with greater motion amplitude and frequency.

The average metric values were found to depend on the condition in which the star drawing task was executed, the type of output kinematic data, and the interaction between these two effects (condition: Pillai's

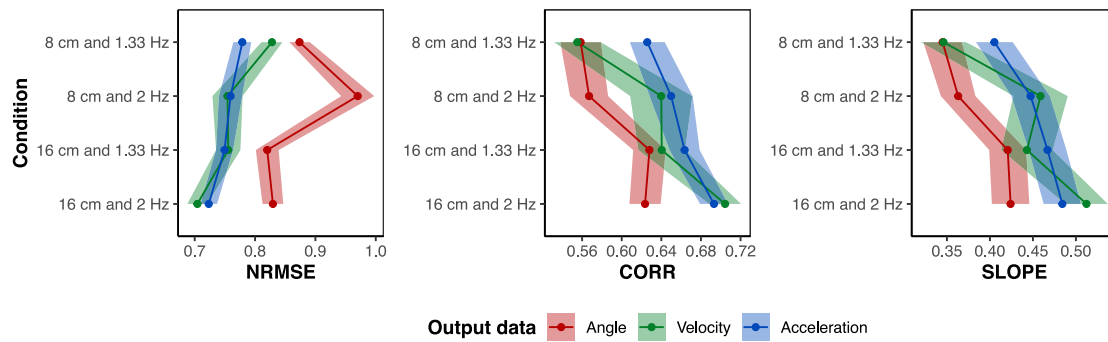


Fig. 5. Performance metric values presented by condition and type of kinematic output data. The points and lines represent the averages computed across the values obtained from all volunteers, joints, and regression models. The shaded areas correspond to the 95% confidence intervals of these averages. In the case of the traditional models, only those trained with their best features are portrayed in the figure. To facilitate visualization, condition averages and confidence intervals were shown using continuous lines, despite the variable being categorical.

Table 8

Overview of the *post-hoc* paired *t*-test results comparing average regression performance between the three types of output kinematic data, with these tests being conducted independently for each condition and performance metric. This table summarizes the obtained outcomes, presenting the total number of tests and the count of statistically significant differences (based on corrected *p*-values). Comparisons between two kinematic data types are separated into two rows: one row for when one of the kinematic types had better average values and another row for when the other kinematic type had better values. Abbreviations: # = number; signif. = significant.

Comparison		NRMSE		CORR		SLOPE	
Better values	Worse values	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests
Acceleration	Velocity	1	2	1	3	0	2
Velocity	Acceleration	0	2	0	1	0	2
Acceleration	Angle	4	4	3	4	2	4
Angle	Acceleration	–	–	–	–	–	–
Velocity	Angle	4	4	1	3	1	4
Angle	Velocity	–	–	0	1	–	–

trace = 1.082, $F = 2.821$, $p = 0.010$; kinematic type: Pillai's trace = 1.352, $F = 6.254$, $p = 0.001$; condition $\times$ kinematic type: Pillai's trace = 1.097, $F = 2.881$, $p < 0.001$). Such interaction effect was also found to be significant when each metric was analyzed using ANOVAs (NRMSE: $F = 7.987$, $p < 0.001$; CORR: $F = 4.016$, $p = 0.005$; SLOPE: $F = 4.535$, $p = 0.003$). Therefore, we conducted paired *t*-tests to investigate the differences between kinematic estimation performances given every condition and metric. The outcomes for these tests are summarized in Table 8. For every condition, joint acceleration estimation led to better metric values than joint angle estimation, with most of these differences being significant. Similarly, velocities were associated with better results than angles for most studied conditions. However, significance was observed primarily for NRMSE. In contrast, when comparing joint accelerations and velocities, no clear pattern was observed, and there were only a few significant *t*-test results (in which accelerations outperformed velocities).

3.5. Kinematic types and joints

The metric values of the different kinematic types and joints are displayed on Fig. 6. Joint accelerations were related to better prediction performances than joint angles. For the shoulder joint, velocities showed better average metric values than the other two kinematic types. For the elbow joint, models predicting accelerations outperformed the others. Additionally, we observed that angle and acceleration prediction were better on average for the elbow joint, whereas velocity prediction was better for the shoulder joint.

The MANOVA indicated that the average metric values were influenced by the type of output kinematic data and the interaction effect between output data type and joint (joint: Pillai's trace = 0.566, $F = 1.307$, $p = 0.416$; kinematic type: Pillai's trace = 1.352, $F = 6.254$, $p = 0.001$; joint $\times$ kinematic type: Pillai's trace = 1.341, $F = 6.100$, $p = 0.001$). The *post-hoc* ANOVAs conducted individually for each metric also supported the presence of this interaction effect (NRMSE:

$F = 23.178$, $p < 0.001$; CORR: $F = 29.580$, $p < 0.001$; SLOPE: $F = 28.429$, $p < 0.001$).

Based on these findings, follow-up paired *t*-tests were performed, and their results are summarized in Table 9. For all metrics and joints, models predicting joint accelerations showed significantly better average performances than those predicting joint angles. In addition, models estimating velocities tended to perform better than those estimating joint angles when considering the number of significant differences. However, no distinct pattern was observed when comparing joint accelerations and velocities.

4. Discussion

We raised the possibility that EMG signals could be more closely related to joint accelerations, followed by joint velocities, and least related to joint angles. Then, we presumed that the closer the association between EMG and kinematic data, the easier it could be for machine learning models to learn such a relationship. This could lead, in turn, to better myoelectric regression performances first when estimating accelerations, then velocities, and finally angles. Therefore, to verify this, our study investigated the performance of EMG regressors in predicting each of these three types of joint kinematic variables. To this end, we conducted a comprehensive comparison, including different regression models, feature extraction methods, conditions, joints, and performance metrics.

In contrast to the prevalent focus on angle estimation in myoelectric control research, our results consistently showed inferior predictive performance for joint angles compared to angular velocities and accelerations. These findings, which aligned with our initial assumptions, suggest that substituting joint angles with alternative kinematic variables as the regression target could improve the overall performance of myoelectric regression models.

Yet, although we found a clear pattern showing worse performances related to angle estimation, it is worth noting that these findings

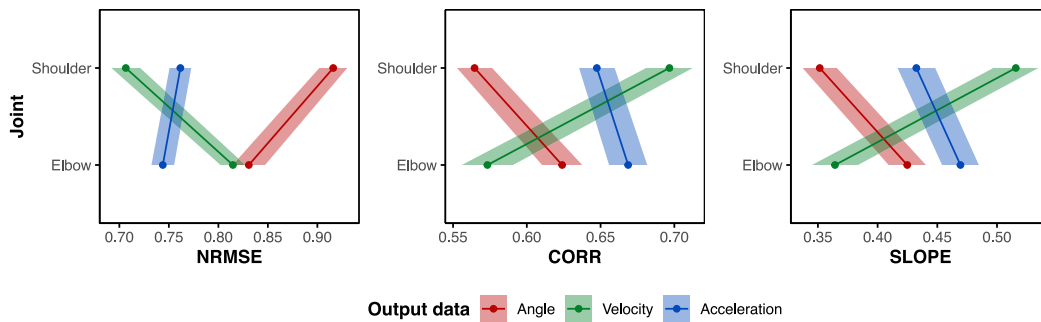


Fig. 6. Performance metric values presented by joint and type of kinematic output data. The points and lines represent the averages computed across the values obtained from all volunteers, conditions, and regression models. The shaded areas correspond to the 95% confidence intervals of these averages. In the case of the traditional models, only those trained with their best features are portrayed in the figure. To facilitate visualization, joint averages and confidence intervals were shown using continuous lines, despite the variable being categorical.

Table 9

Overview of the *post-hoc* paired *t*-test results comparing average regression performance between the three types of output kinematic data, with these tests being conducted independently for each joint and performance metric. This table summarizes the obtained outcomes, presenting the total number of tests and the count of statistically significant differences (based on corrected *p*-values). Comparisons between two kinematic data types are separated into two rows: one row for when one of the kinematic types had better average values and another row for when the other kinematic type had better values. Abbreviations: # = number; signif. = significant.

Comparison		NRMSE		CORR		SLOPE	
Better values	Worse values	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests	# of signif. tests	Total # of tests
Acceleration	Velocity	1	1	1	1	1	1
Velocity	Acceleration	1	1	0	1	1	1
Acceleration	Angle	2	2	2	2	2	2
Angle	Acceleration	–	–	–	–	–	–
Velocity	Angle	1	2	1	1	1	1
Angle	Velocity	–	–	0	1	1	1

diverge from some other results in the literature. Whereas, to the best of our knowledge, our article is the first to focus on differences in regression performance for distinct joint kinematic variables, there are a few other myoelectric control studies that have also used more than one kinematic variable, providing a basis for comparison with our results.

Hajian and Morin (2022), for instance, proposed a two-stream CNN to simultaneously estimate elbow joint angles and velocities from raw EMG signals. The data used in their study were recorded during elbow flexion and extension movements on the sagittal plane. These movements were performed under three different conditions: with controlled joint angular velocity, with controlled joint torque, and free movements without control. The authors found better velocity estimation during movements with controlled angular velocity. However, in the case of free movements, angles displayed superior results. When torque was controlled, angles were again better than velocities, but without statistically significant differences reported between the two kinematic variables. Huang, Chen, Zhang, Wang, and Ota (2021) investigated the use of MLPs fed with an EMG feature processed through a continuous wavelet transform for estimating either elbow extension-flexion joint angle or velocity. Their experimental procedure consisted of moving back and forth a robotic arm in the sagittal plane given three different movement directions (parallel, perpendicular, and diagonal to the ground) and three different resistances offered by the device (low, medium, and high). Their study revealed better performance in velocity estimation for motions parallel to the ground under medium and high resistances, whereas angles performed better in the other conditions. Nasr et al. (2021) proposed a procedure using machine learning models to predict shoulder joint elevation torques, angular positions, velocities, or accelerations. They investigated arm movements in the sagittal plane consisting of a pick-and-place task of moving a bottle. Their experiments found angles to outperform torques, then velocities, and lastly accelerations in regression performance. Finally, Raj and Sivanandan (2017) studied elbow extension and flexion motions in the sagittal plane using varying angular velocities. While their article used neural

network models to simultaneously predict joint angles and velocities, the predicted velocities were quantized into average velocities. Therefore, their approach differs from ours and the previous three studies as they did not aim to map EMG to instantaneous continuous velocity. Hence, in the following paragraphs, we will focus on comparing our results to those of the first three other articles.

One possible explanation for these differing results may stem from the overall properties of the motion signal waveforms obtained in each study.

While the primary focus of Hajian and Morin (2022), Huang et al. (2021), and Nasr et al. (2021) works was away from specifically contrasting differences between joint kinematics, their papers included figures with examples of different motion signals, offering us a window to analyze their waveforms. In Hajian and Morin (2022), it seems that velocities obtained during movements with controlled angular velocities exhibited a relatively simple trapezoidal-like waveform with consistent standard deviations. For freely executed movements, the velocity waveforms appear to have more intricate patterns, with sharp peaks and troughs, plateaus around zero values, and more varying standard deviations. When torques were controlled, the velocity waveforms seemed to show again these more intricate patterns but with more constant standard deviations. In contrast, angles across all movement types in their study seem to present relatively simple sinusoidal-like waveforms with constant standard deviations. In the study by Huang et al. (2021), it appears that angular velocities had more irregular waveforms, with sharper peaks and troughs, and more high-frequency variations. Angles, in their case, seem to have a simpler sinusoidal-like pattern with less high-frequency variations. Lastly, the work of Nasr et al. (2021) provided examples of joint angles and torque signals. According to these examples, it seems that torques presented more irregular waveforms, with some high-frequency variations. Similar to the other studies, angles appear to exhibit simpler waveforms with fewer high-frequency variations and constant standard deviations.

A representative example of the joint angular positions, velocities, and accelerations observed in our study is displayed in Fig. 7. In

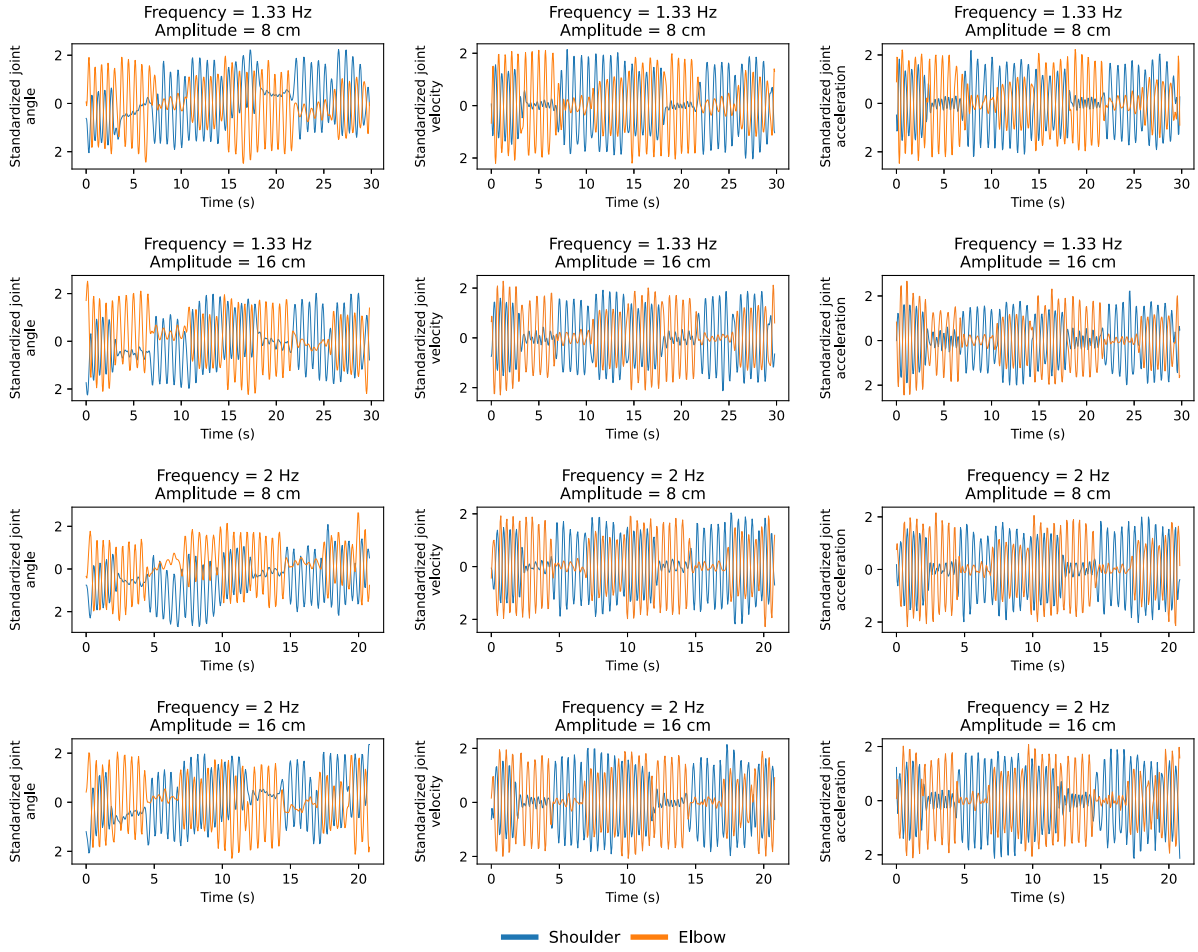


Fig. 7. Representative example of the obtained joint kinematics. Standardized joint angular positions, velocities, and accelerations are shown for trials performed with different motion frequencies and amplitudes. Kinematics for both shoulder and elbow joints from one volunteer are displayed.

our data, all kinematic variables generally resembled sine waves with constant frequencies throughout each trial, albeit with varying standard deviations. These varying amplitudes within a trial relate to the four distinct line orientations of the star shape that volunteers were instructed to trace. Such variations are due to the different demands placed on the shoulder and elbow joints when executing movements in distinct directions. For joint angular velocities and accelerations, the moving average value seemed to be relatively constant across each trial. In contrast, joint angles displayed occasional shifts in their moving average values within trials, adding a relative layer of unpredictability to the angle waveforms in our study.

Given a fixed EMG input signal X , a more intricate output signal Y could make it harder for machine learning algorithms to model the input–output relationship, leading to worse regression estimates. This is because such intricacy could manifest as multiple modes or high variance in the conditional probability distribution of $P(Y|X)$, possibly resulting in several Y values having similarly high probabilities for a given X . In other words, different joint kinematic values having similar probabilities given a specific EMG pattern. This increased uncertainty complicates finding the Y value that maximizes $P(Y|X)$, which is often the objective of regression tasks. Moreover, this intricacy could introduce ambiguity in the input–output relationship by making the distributions of $P(X|Y)$ less distinguishable between multiple Y values. That is, a specific joint kinematic variable value could be achieved by various EMG patterns. Hence, this ambiguity could further hinder the models' learning capability.

Perhaps these could be contributing factors in our results, as well as those from [Hajian and Morin \(2022\)](#), [Huang et al. \(2021\)](#), and [Nasr](#)

[et al. \(2021\)](#). For instance, in [Hajian and Morin \(2022\)](#), angles yielded better estimates when velocity waveforms appeared more intricate. However, velocities surpassed angles when both waveform types were relatively simple. In [Huang et al. \(2021\)](#), angle prediction showed superior results in most cases, while their angle waveforms appeared to be relatively simpler in all cases. [Nasr et al. \(2021\)](#) also reported superior angle predictions when compared to torques, while the waveforms of the former were apparently simpler than those of the latter. In our study, angle waveforms exhibited additional unpredictability due to shifts in their moving average values, which could be a factor contributing to their worse regression performance. Therefore, this apparent intricateness and unpredictability may serve as an alternative or supplementary explanation for our findings, alongside the less direct relationship between EMG and joint angles initially suggested by us.

Given the potential impact of output signal waveform properties on myoelectric regression performance, we advise future research to consider this factor in their experimental design. Specifically, the choice of movements and methods for kinematic signal processing to be studied may be important aspects influencing these waveforms, and, consequently, may be relevant to enhancing myoelectric device performance. To further deepen the field's understanding of this area, future studies could investigate how changes in output waveform properties impact the regression estimates for various kinematic variables. This could be accomplished, for example, by exploring different types of movements or multiple techniques for recording and processing motion data.

Moreover, another possible reason for the differing results among studies could be related to the orientation of the movements chosen for each study. In our dataset, motions were predominantly horizontal,

with only a minor vertical component. In contrast, most movements in the three other articles occurred in the sagittal plane, with a considerable vertical component. The exception was one motion direction used in Huang et al. (2021), which involved horizontal movements parallel to the ground. It was observed superior estimations of velocities rather than angles for the horizontal movements analyzed by us and for most of those studied by Huang et al. (2021). Conversely, the more vertical movements in the previous three studies often yielded better angle estimations. The vertical direction of these movements introduces a varying gravitational effect on the arm throughout the motions due to changes in limb orientation relative to the gravity field. Such a varying effect could lead to more intricate muscle activation patterns throughout the motions. Perhaps this could distinctly affect the machine learning algorithms' ability to predict different kinematic variables. While it is still unclear if such a distinct effect exists, this topic may present an interesting avenue for future research.

Another finding from our article is that, while accelerations showed slightly better average performances than velocities, most of the differences between the two were not statistically significant. A few factors might be behind these results. First, EMG signals could be more directly related to the way muscles are contracting rather than the current joint state. Angular accelerations and velocities are variables that better represent joint movement. On the other hand, angles portray more clearly the joint state. As both accelerations and velocities more closely represent the movement as captured by EMG signals, the difference in regression performance in predicting both might not be as strong as the difference between the two variables and joint angles. Second, the waveform properties of both accelerations and velocities in our dataset appeared quite similar, reducing then the likelihood of this aspect causing noticeable performance differences between them. Lastly, our kinematic data were derived in the following order: from 3D motion capture markers' positions to joint angles, then to velocities, and finally to accelerations. Hence, joint kinematics were not directly measured; they were obtained from other variables in a sequential manner. This sequential process could inherently lead to accumulating measurement errors as the derivation process proceeds. While the quantification of these errors is not possible within our study (as the direct joint kinematic measurements are not available), it is possible that these errors could have blurred a performance difference between accelerations and velocities, especially if this difference was not large. Future research could either explore alternative preprocessing methods for deriving kinematic variables or direct measurements of these variables, shedding light on the impact of such error accumulation on joint kinematic regression.

Finally, other aspects that researchers and practitioners should be attentive to when selecting the best target joint kinematic variable for their future applications are the motion signal preprocessing, hyperparameter optimization, and training procedures, as well as the evaluation metrics. For example, when a model optimization (either hyperparameter optimization or model training) takes into account more than one kinematic variable, the units of these variables and the optimized metric should be carefully considered. These variables have inherently different units and likely will have different value ranges. When non-dimensionless metrics are optimized, such as the commonly used MSE, the kinematic variables with larger value magnitudes will likely produce larger values for these metrics. Therefore, the optimization process might show an undesired tendency to select model parameters or hyperparameters that will yield more favorable results for the kinematic variable with larger value magnitudes. A potential solution for this issue is to rescale all the kinematic variables to be in a similar unitless scale before model hyperparameter optimization and training, as we did in this paper. This ensures that the optimized metrics are more comparable between the different variables. Likewise, when comparing the regression results of different kinematic variables, it may be more meaningful to use dimensionless metrics, such as those employed here. Moreover, we encourage future studies to provide detailed descriptions of the hyperparameter optimization and training procedures since this could help verify the absence of biases towards a certain variable during optimization and improve reproducibility.

5. Conclusion

In summary, this paper studied the performance of myoelectric regressors when predicting different types of joint kinematic variables. A comprehensive comparison was conducted, including different regression models, feature extraction methods, conditions, joints, and performance metrics. Contrary to the predominant practice of employing joint angles as the target kinematic variable of EMG regressors, our findings show consistently superior regression performances when estimating joint angular accelerations and velocities rather than joint angles. This observation was discussed in-depth in the context of current literature, and several factors that could be contributing to our findings were proposed, including: aspects of the relationship between different kinematic variables and the EMG signals, the waveform properties of the obtained kinematic variables, the characteristics of the analyzed movement, and an intrinsic error accumulation when deriving these variables. Based on this discussion, recommendations were made for future works, and areas for future research were suggested. Overall, these insights challenge the default use of joint angles and could pave the way for more effective myoelectric assistive technologies, thereby enhancing the quality of life for end-users.

CRediT authorship contribution statement

Eric Cito Becman: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration, Funding acquisition. **Larissa Driemeier:** Writing – review & editing, Supervision, Project administration. **Oron Levin:** Methodology, Investigation, Data curation. **Stephan P. Swinnen:** Resources, Supervision, Project administration. **Arturo Forner-Cordero:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was collected for the research described in the article.

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Appendix A. Review on the use of kinematics for myoelectric control

A literature review was conducted to examine the frequency of using each type of joint kinematic data (joint angle, velocity, and acceleration) as the output for EMG-based regression systems. To achieve this, we searched the Scopus bibliographic database for peer-reviewed journal articles in English containing original research on the subject. We conducted three separate search queries, one for each of the three kinematic data types under consideration. Each query was designed to retrieve articles that discussed topics related to EMG, regression, and the respective kinematic type. To accomplish this, we searched for relevant terms within the titles, abstracts, and keywords of the articles in the database. Next, we examined the titles and abstracts for each article to determine if they met our selection criteria. If uncertainty remained after reviewing the title and abstract, we proceeded to inspect the full article text. Upon completing the manual evaluation, we finished with a

list of 91 articles that used EMG for joint angle regression, 4 articles that employed EMG for joint velocity regression, and 2 articles that applied EMG for joint acceleration regression (for improved readability, this list has been placed in the Supplementary Materials). These numbers indicate that using joint angles in myoelectric control is considerably more common than the other joint kinematic types under consideration.

Appendix B. Adapted feature extraction methods

In Khushaba et al. (2017), the CV feature (which they refer to as COV) is defined as the standard deviation of a time series divided by its mean. The CV is also used to obtain the 6th feature of the TDD feature extractor (called TDD [6/7] in our article and f_6 in theirs), which is equal to the logarithm of the CV. Here, we make a small modification to the CV definition, and, consequently, to the TDD feature extractor as well. Instead of dividing by the mean, we divide by the absolute mean. This simple modification ensures that CV is not negative, making it a closer representation of a relative standard deviation (which cannot be negative) and avoiding the computation of logarithms of negative values.

Also, in Khushaba et al. (2017), the authors transform the EMG signal using the Teager–Kaiser energy operator, sum all of its values, and then compute the 7th feature of the TDD feature extractor (named TDD [7/7] in our article and f_7 in theirs) as the logarithm of this sum. In our article, we modified this procedure by computing the sum over the absolute values of the Teager–Kaiser energy operator, which we refer to as ITKEO. This ensures that negative values do not partially cancel out positive transformed values during summation, and also avoids the possibility of calculating the logarithm of negative values in the TDD feature extractor.

The MMAV3 feature is a modification we propose to the MMAV2 feature. The definition of the latter is the following:

$$\text{MMAV2} = \frac{1}{N} \sum_{i=1}^N w_i |x_i|;$$

$$w_i = \begin{cases} 1, & \text{if } 0.25N \leq i \leq 0.75N \\ 4i/N, & \text{if } i < 0.25N \\ 4(N-i)/N, & \text{otherwise} \end{cases} \quad (1)$$

In the formula above, N is the length of the EMG time series, x_i is the signal value at time point i , and w_i is a weight value at the same time point. Therefore, it is a weighted version of the MAV feature function. However, as i is always less or equal to N , the w_i is always negative or zero for $i > 0.75N$. In practice, this means that the positive weighted absolute values from the first 3 quarters of the signal will be partially canceled out during summation by the negative weighted values from the last quarter. This goes against the rationale behind the MAV function, in which all time points positively contribute towards the final value. To avoid this and to create a smoother weighted version of the MAV, we propose the MMAV3 as a simple modification of the MMAV2, as in:

$$\text{MMAV3} = \frac{1}{N} \sum_{i=1}^N w_i |x_i|;$$

$$w_i = \begin{cases} 1, & \text{if } 0.25N \leq i \leq 0.75N \\ 4i/N, & \text{if } i < 0.25N \\ 4(N-i)/N, & \text{otherwise} \end{cases} \quad (2)$$

By replacing $4(N-i)/N$ with $4(N-i)/N$, we avoid negative w_i values.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.eswa.2024.124345>.

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